

Integumentary System

1. Which of the following is composed of loose connective tissue?

- a. Epidermis
- b. Reticular layer of dermis
- c. Hypodermis
- d. papillary layer of dermis
- e. Both b and c

Answer: c

The skin is composed of two layers: the epidermis and the dermis. Underneath these layers lies the hypodermis. The epidermis is stratified squamous epithelium. The dermis is composed of a papillary layer and a reticular layer. The reticular layer of the dermis is made up of dense irregular connective tissue. The hypodermis is a layer of loose connective tissue.

2.

Where is thick skin found?

- a. Over the knee
- b. Sole of the feet
- c. Breast
- d. Lips
- e. All of the above

Answer: b

3.

Which layer of the epidermis has cells which have keratohyaline granules?

- a. Stratum basale
- b. Stratum spinosum
- c. Stratum granulosum
- d. Stratum lucidum
- e. Stratum corneum

Answer: c

The epidermis is divided into five layers: stratum basale, stratum spinosum, stratum granulosum, stratum lucidum, and stratum corneum.

The stratum basale contains the dividing cells. This layer is also called the stratum germinativum.

The stratum spinosum consists of a layer several cells deep. The cells have pointy or spiny processes on them. The cells in the stratum granulosum contain keratohyaline granules.

The stratum lucidum is present only in thick skin.

The stratum corneum is the outermost layer. The cells in this layer are essentially bags of keratin. They contain no nuclei or organelles.

Histology hint from Sarah Bellham: My favorite mnemonic for remembering the layers of the skin: "Californian Ladies Give Superb Backrubs"

4.

Which cell is a macrophage found in the skin?

- a. Kupffer cells
- b. Histiocyte
- c. Dust cell
- d. Langerhans cell
- e. Microglia

Answer: d

Macrophages are mononuclear phagocytes. Many tissues have resident (fixed) macrophages. Fixed macrophages are given a unique name, depending on the tissue that they are located in. Kupffer cells are the hepatic macrophages. Histiocytes are macrophages seen in connective tissue. Dust cells are alveolar macrophage found in the respiratory tract. Langerhans cells are macrophages seen in the skin. Microglia are the central nervous system macrophages.

5.

A new miracle skin cream recently hit the beauty counters which is suppose to stimulate collagen production. Which cell is it supposedly stimulating?

- a. Langerhans cell
- b. Keratinocyte
- c. Melanocyte
- d. Merkel cell
- e. Fibroblast

Answer: e

Langerhans cells, keratinocytes, melanocytes, and Merkel cells are all found in the epidermis. The Langerhans cell is a phagocyte. The keratinocyte is the most abundant cell in the epidermis. The melanocyte produces melanin, which is responsible for skin pigmentation. The Merkel cell is a mechanoreceptor.

Fibroblasts are found in the dermis. Fibroblasts produces collagen.

6.

What is the half moon shaped white area on a nail called?

- a. Lunula
- b. Eponychium
- c. Matrix
- d. Nail bed
- e. Root

Answer: a

The lunula is the half moon shaped white area on a nail. The anatomical term for the cuticle is the eponychium. The matrix is the region of the nails where there are dividing cells and nail growth. The nail plate rests on the nail bed. The nail root is the proximal portion of the nail that is underneath skin.

7.

What is the growing part of the nail?

- a. Lunula
- b. Eponychium
- c. Matrix
- d. Nail bed
- e. Root

Answer: c

The lunula is the half moon shaped white area on a nail. The anatomical term for the cuticle is the eponychium. The matrix is the region of the nails where there are dividing cells and nail growth. The nail plate rests on the nail bed. The nail root is the proximal portion of the nail that is underneath skin.

8.

What type of glands are the ceruminous glands?

- a. Sebaceous glands
- b. Eccrine sweat gland
- c. Endocrine gland
- d. Apocrine sweat gland
- e. Oil gland

Answer: d

The ceruminous glands of the ear are apocrine sweat glands.

9.

Which of the following is the most abundant sensory receptor of the skin?

- a. Free nerve endings
- b. Ruffini's corpuscles
- c. Pacinian corpuscles

- d. Krause's end bulbs
- e. Meissner's corpuscle

Answer: a

There are several different sensory receptors in the skin. The most abundant sensory receptor are the free nerve endings. Free nerve endings respond to pain and temperature. Ruffini's corpuscles respond to continuous pressure. Pacinian corpuscles respond to vibration and rapidly changing pressure. Krause's end bulbs are a receptor for fine touch which are located in mucous membranes and the tongue. Meissner's corpuscles are also a receptor for fine touch but they are located in the dermis.

10.

Where can hair be found?

- a. Palms of hand
- b. Soles of feet
- c. Urogenital openings
- d. Lips
- e. Eyelid

Hair is present over most of the body. It is not found on the palms of the hand, soles of the feet, urogenital openings, and lips.

1.

Which of the following is composed of connective tissue?

- a. Epidermis
- b. Dermis
- c. Hypodermis
- d. Both a and b
- e. Both b and c

Answer: e

The skin is composed of two layers: the epidermis and the dermis. Underneath these layers lies the hypodermis. The epidermis is stratified squamous epithelium. The dermis is composed of a papillary layer and a reticular layer. The reticular layer of the dermis is made up of dense irregular connective tissue. The hypodermis is a layer of loose connective tissue.

2.

Which of the following is composed of dense irregular connective tissue?

- a. Epidermis
- b. Reticular layer of dermis
- c. Hypodermis
- d. Both a and b
- e. Both b and c

Answer: b

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3.

Where is thick skin found?

- a. Lips
- b. Over the knee
- c. Palms
- d. Breast
- e. All of the above

Answer: c

Thick skin is found on the palms of the hand and the sole of the feet. Thin skin is found everywhere else.

4.

Which layer of the epidermis is present only in thick skin?

- a. Stratum basale
- b. Stratum spinosum
- c. Stratum granulosum
- d. Stratum lucidum
- e. Stratum corneum

Answer: d

The epidermis is divided into five layers: stratum basale, stratum spinosum, stratum granulosum, stratum lucidum, and stratum corneum.

The stratum basale contains the dividing cells. This layer is also called the stratum germinativum.

The stratum spinosum consists of a layer several cells deep. The cells have pointy or spiny processes on them. The cells in the stratum granulosum contain keratohyaline granules.

The stratum lucidum is present only in thick skin.

The stratum corneum is the outermost layer. The cells in this layer are essentially bags of keratin. They contain no nuclei or organelles.

Note from Sarah Bellham: My favorite mnemonic for remembering the layers of the skin: "Californian Ladies Give Superb Backrubs"

5.

Which cell is a mechanoreceptors?

- a. Langerhans cell
- b. Keratinocyte
- c. Melanocyte
- d. Merkel cell
- e. Fibroblast

Answer: d

Langerhans cells, keratinocytes, melanocytes, and Merkel cells are all found in the epidermis. The Langerhans cell is a phagocyte. The keratinocyte is the most abundant cell in the epidermis. The melanocyte produces melanin, which is responsible for skin pigmentation. The Merkel cell is a mechanoreceptor.

Fibroblasts are found in the dermis. Fibroblasts produces collagen.

6.

Which cell is found in the dermis?

- a. Langerhans cell
- b. Keratinocyte
- c. Melanocyte
- d. Merkel cell
- e. Fibroblast

Answer: e

Langerhans cells, keratinocytes, melanocytes, and Merkel cells are all found in the epidermis. The Langerhans cell is a phagocyte. The keratinocyte is the most abundant cell in the epidermis. The melanocyte produces melanin, which is responsible for skin pigmentation. The Merkel cell is a mechanoreceptor.

Fibroblasts are found in the dermis. Fibroblasts produces collagen.

7.

What is underneath the nail plate?

- a. Lunula
- b. Eponychium
- c. Matrix
- d. Nail bed
- e. Root

Answer: d

The lunula is the half moon shaped white area on a nail. The anatomical term for the cuticle is the eponychium. The matrix is the region of the nails where there are dividing cells and nail growth. The nail plate rests on the nail bed. The nail root is the proximal portion of the nail that is underneath skin.

8.

What is the innervation of an eccrine sweat gland?

- a. Cholinergic; parasympathetic
- b. Cholinergic; sympathetic
- c. Adrenergic; parasympathetic
- d. Adrenergic; sympathetic
- e. Cholinergic; motor

Answer: b

Eccrine sweat glands are innervated by the sympathetic nervous system. The neurotransmitter for the eccrine sweat glands is acetylcholine. Thus it is cholinergic.

Note from Sarah Bellham: For most postganglionic sympathetic neurons, the neurotransmitter is norepinephrine. Eccrine sweat glands are an exception to this generalization, as the innervation for eccrine sweat glands is cholinergic sympathetic.

9.

Which of the following responds to continuous pressure?

- a. Free nerve endings
- b. Ruffini's corpuscles
- c. Pacinian corpuscles
- d. Krause's end bulbs
- e. Meissner's corpuscle

Answer: b

There are several different sensory receptors in the skin. The most abundant sensory receptor are the free nerve endings. Free nerve endings respond to pain and temperature. Ruffini's corpuscles respond to continuous pressure. Pacinian corpuscles respond to vibration and rapidly changing pressure. Krause's end bulbs are a receptor for fine touch which are located in mucous membranes and the tongue. Meissner's corpuscles are also a receptor for fine touch but they are located in the dermis.

10.

Which of the following is found in the hair follicle?

- a. Pigment epithelium
- b. Muller's cells
- c. Huxley's layer
- d. Horizontal cells
- e. Cones

Answer: c

Pigment epithelium, Muller's cells, horizontal cells, and cones are all part of the retina.

Huxley's layer is a layer in the hair follicle.

1.

The reticular layer is part of which layer?

- a. Epidermis
- b. Dermis
- c. Hypodermis
- d. Both a and b
- e. None of the above

Answer: b

The skin is composed of two layers: the epidermis and the dermis. Underneath these layers lies the hypodermis. The epidermis is stratified squamous epithelium. The dermis is composed of a papillary layer and a reticular layer. The reticular layer of the dermis is made up of dense irregular connective tissue. The hypodermis is a layer of loose connective tissue.

2.

Which of the following is NOT considered an epidermal appendage?

- a. Sweat gland
- b. Hair
- c. Hypodermis
- d. Nails
- e. Sebaceous glands

Answer: c

Sweat glands, hair, nails and sebaceous glands are all considered epidermal appendages. The hypodermis is not considered an epidermal appendage. The hypodermis is the loose connective tissue layer underneath the dermis.

3.

Which layer of the epidermis is also called the stratum germinativum?

- a. Stratum basale
- b. Stratum spinosum
- c. Stratum granulosum
- d. Stratum lucidum
- e. Stratum corneum

Answer: a

The epidermis is divided into five layers: stratum basale, stratum spinosum, stratum granulosum, stratum lucidum, and stratum corneum.

The stratum basale contains the dividing cells. This layer is also called the stratum germinativum.

The stratum spinosum consists of a layer several cells deep. The cells have pointy or spiny processes on them. The cells in the stratum granulosum contain keratohyaline granules.

The stratum lucidum is present only in thick skin.

The stratum corneum is the outermost layer. The cells in this layer are essentially bags of keratin. They contain no nuclei or organelles.

Note from Sarah Bellham: My favorite mnemonic for remembering the layers of the skin: "Californian Ladies Give Superb Backrubs"

4.

Which layer of the epidermis is on the surface of the skin?

- a. Stratum basale
- b. Stratum spinosum
- c. Stratum granulosum
- d. Stratum lucidum
- e. Stratum corneum

Answer: e

The epidermis is divided into five layers: stratum basale, stratum spinosum, stratum granulosum, stratum lucidum, and stratum corneum.

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5.

Which cell is the most abundant cell in the epidermis?

- a. Langerhans cell

- b. Keratinocyte
- c. Melanocyte
- d. Merkel cell
- e. Fibroblast

Answer: b

Langerhans cells, keratinocytes, melanocytes, and Merkel cells are all found in the epidermis. The Langerhans cell is a phagocyte. The keratinocyte is the most abundant cell in the epidermis. The melanocyte produces melanin, which is responsible for skin pigmentation. The Merkel cell is a mechanoreceptor.

Fibroblasts are found in the dermis. Fibroblasts produces collagen.

6.

What type of epithelium forms the epidermis?

- a. Simple squamous epithelium
- b. Simple cuboidal epithelium
- c. Simple columnar epithelium
- d. Stratified squamous epithelium
- e. Pseudostratified epithelium

Answer: d

The epidermis is formed by stratified squamous epithelium. Epithelium lines body cavities and surfaces. Stratified squamous epithelium is "stratified" because it is more than one cell layer thick. "Squamous" refers to the fact that the surface cells of the stratified squamous layer are flat. This is important to note as the cells at the basal layer are cuboidal or even columnar. It is still classified as "squamous" based on the cells of the surface layer.

7.

What is the portion of the nail which is underneath skin?

- a. Lunula
- b. Eponychium
- c. Matrix
- d. Nail bed
- e. Root

Answer: e

The lunula is the half moon shaped white area on a nail. The anatomical term for the cuticle is the eponychium. The matrix is the region of the nails where there are dividing cells and nail growth. The nail plate rests on the nail bed. The nail root is the proximal portion of the nail that is underneath skin.

8.

What type of glands are the glands of Moll?

- a. Endocrine gland
- b. Apocrine sweat gland
- c. Oil gland
- d. Sebaceous glands
- e. Eccrine sweat gland

Answer: b

The glands of Moll in the eyelid are apocrine sweat glands.

9.

Which of the following responds to vibration and rapidly changing pressure?

- a. Free nerve endings
- b. Ruffini's corpuscles
- c. Pacinian corpuscles
- d. Krause's end bulbs
- e. Meissner's corpuscle

Answer: c

There are several different sensory receptors in the skin. The most abundant sensory receptor are the free nerve endings. Free nerve endings respond to pain and temperature. Ruffini's corpuscles respond to continuous pressure. Pacinian corpuscles respond to vibration and rapidly changing pressure. Krause's end bulbs are a receptor for fine touch which are located in mucous membranes and the tongue. Meissner's corpuscles are also a receptor for fine touch but they are located in the dermis.

10.

Which of the following is a receptor for fine touch which is located in mucous membranes?

- a. Free nerve endings
- b. Ruffini's corpuscles
- c. Pacinian corpuscles
- d. Krause's end bulbs
- e. Meissner's corpuscle

Answer: d

There are several different sensory receptors in the skin. The most abundant sensory receptor are the free nerve endings. Free nerve endings respond to pain and temperature. Ruffini's corpuscles respond to continuous pressure. Pacinian corpuscles respond to vibration and rapidly changing pressure. Krause's end bulbs are a receptor for fine touch which are located in mucous membranes and the tongue. Meissner's corpuscles are also a receptor for fine touch but they are located in the dermis.

1.

Which of the following is composed of stratified squamous epithelium?

- a. Epidermis
- b. Dermis
- c. Hypodermis
- d. Both a and b
- e. Both b and c

Answer: a

The skin is composed of two layers: the epidermis and the dermis. Underneath these layers lies the hypodermis. The epidermis is stratified squamous epithelium. The dermis is composed of a papillary layer and a reticular layer. The reticular layer of the dermis is made up of dense irregular connective tissue. The hypodermis is a layer of loose connective tissue.

2.

Which of the following is NOT a function of skin

- a. Vitamin D production
- b. Protection from water loss
- c. Sensory reception
- d. Heat regulation
- e. All are functions of skin

Answer: e

The skin is involved in the production of vitamin D from precursors with the aid of the sun. It protects the body from water loss. There are many sensory receptors in the skin: pain, pressure, fine touch. The skin is also involved in heat regulation. In addition, the skin protects the body.

3.

Which layer of the epidermis contains dividing cells?

- a. Stratum basale
- b. Stratum spinosum
- c. Stratum granulosum
- d. Stratum lucidum
- e. Stratum corneum

Answer: a

The epidermis is divided into five layers: stratum basale, stratum spinosum, stratum granulosum, stratum lucidum, and stratum corneum.

The stratum basale contains the dividing cells. This layer is also called the stratum germinativum.

The stratum spinosum consists of a layer several cells deep. The cells have pointy or spiny processes on them. The cells in the stratum granulosum contain keratohyaline granules.

The stratum lucidum is present only in thick skin.

The stratum corneum is the outermost layer. The cells in this layer are essentially bags of keratin. They contain no nuclei or organelles.

Note from Sarah Bellham: My favorite mnemonic for remembering the layers of the skin: "Californian Ladies Give Superb Backrubs"

4.

Which layer of the epidermis contains cells with no nuclei or organelles?

- a. Stratum basale
- b. Stratum spinosum
- c. Stratum granulosum
- d. Stratum lucidum
- e. Stratum corneum

Answer: e

The epidermis is divided into five layers: stratum basale, stratum spinosum, stratum granulosum, stratum lucidum, and stratum corneum.

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5.

Which cell is a phagocyte?

- a. Langerhans cell
- b. Keratinocyte
- c. Melanocyte
- d. Merkel cell
- e. Fibroblast

Answer: a

Langerhans cells, keratinocytes, melanocytes, and Merkel cells are all found in the epidermis. The Langerhans cell is a phagocyte. The keratinocyte is the most abundant cell in the epidermis. The melanocyte produces melanin, which is responsible for skin pigmentation. The Merkel cell is a mechanoreceptor.

Fibroblasts are found in the dermis. Fibroblasts produces collagen.

6.

What is a characteristic of the cells in the epidermis of the skin?

- a. Microvilli
- b. Stereocilia
- c. Cilia
- d. Keratinization
- e. Both a and b

Answer: d

Microvilli are the finger like projections seen on the surface of some cells. The appearance of microvilli form what is also called the brush border or striated border.

Stereocilia are very long microvilli. Stereocilia are seen in the epididymis and the hair cells of the ear.

Cilia is the hair like surface modification seen on some epithelia. Cilia are made of microtubules.

Keratinization is seen in the epidermis. Cells in the stratum corneum are essentially just bags of keratin.

7.

Where are apocrine sweat glands NOT found?

- a. Areola
- b. External genitalia
- c. Posterior neck
- d. Axilla
- e. Circumanal region

Answer: c

Sweat glands are divided into apocrine and eccrine. Apocrine sweat glands are found on the areola, external genitalia, axilla, and circumanal region. Eccrine sweat glands are distributed over most of the body.

8.

What is the innervation of an apocrine sweat gland?

- a. Cholinergic; parasympathetic
- b. Cholinergic; sympathetic
- c. Adrenergic; parasympathetic

- d. Adrenergic; sympathetic
- e. Cholinergic; motor

Answer: d

Apocrine sweat glands are innervated by the sympathetic nervous system. The neurotransmitter for the apocrine sweat glands is norepinephrine. Thus it is adrenergic.

9.

Which of the following is a receptor for fine touch which is located in the dermis?

- a. Free nerve endings
- b. Ruffini's corpuscles
- c. Pacinian corpuscles
- d. Krause's end bulbs
- e. Meissner's corpuscle

Answer: e

There are several different sensory receptors in the skin. The most abundant sensory receptor are the free nerve endings. Free nerve endings respond to pain and temperature. Ruffini's corpuscles respond to continuous pressure. Pacinian corpuscles respond to vibration and rapidly changing pressure. Krause's end bulbs are a receptor for fine touch which are located in mucous membranes and the tongue. Meissner's corpuscles are also a receptor for fine touch but they are located in the dermis.

10.

Which of the following is found in the hair follicle?

- a. Henle's layer
- b. Bipolar cells
- c. Amacrine cells
- d. Rods
- e. Ganglion cells

Answer: a

Bipolar cells, amacrine cells, rods, and ganglion cells are all part of the retina.

Henle's layer is a layer in the hair follicle.

1.

The papillary layer is part of which layer?

- a. Epidermis
- b. Dermis
- c. Hypodermis
- d. Both a and b
- e. None of the above

Answer: b

The skin is composed of two layers: the epidermis and the dermis. Underneath these layers lies the hypodermis. The epidermis is stratified squamous epithelium. The dermis is composed of a papillary layer and a reticular layer. The reticular layer of the dermis is made up of dense irregular connective tissue. The hypodermis is a layer of loose connective tissue.

2.

Which layer of the epidermis contains star shaped cells?

- a. Stratum basale
- b. Stratum spinosum
- c. Stratum granulosum
- d. Stratum lucidum
- e. Stratum corneum

Answer: b

The epidermis is divided into five layers: stratum basale, stratum spinosum, stratum granulosum, stratum lucidum, and stratum corneum.

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The stratum lucidum is present only in thick skin.

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Note from Sarah Bellham: My favorite mnemonic for remembering the layers of the skin: "Californian Ladies Give Superb Backrubs"

3.

Which of the following layers comprise the skin?

- a. Epidermis
- b. Dermis
- c. Hypodermis
- d. Both a and b
- e. Both b and c

Answer: d

The skin is composed of two layers: the epidermis and the dermis. Underneath these layers lies the hypodermis. The epidermis is stratified squamous epithelium. The dermis is composed of a papillary

layer and a reticular layer. The reticular layer of the dermis is made up of dense irregular connective tissue. The hypodermis is a layer of loose connective tissue.

4.

Which cell is responsible for skin pigmentation?

- a. Langerhans cell
- b. Keratinocyte
- c. Melanocyte
- d. Merkel cell
- e. Fibroblast

Answer: c

Langerhans cells, keratinocytes, melanocytes, and Merkel cells are all found in the epidermis. The Langerhans cell is a phagocyte. The keratinocyte is the most abundant cell in the epidermis. The melanocyte produces melanin, which is responsible for skin pigmentation. The Merkel cell is a mechanoreceptor.

Fibroblasts are found in the dermis. Fibroblasts produces collagen.

5.

Which of the following responds to pain?

- a. Free nerve endings
- b. Ruffini's corpuscles
- c. Pacinian corpuscles
- d. Krause's end bulbs
- e. Meissner's corpuscle

Answer: a

There are several different sensory receptors in the skin. The most abundant sensory receptor are the free nerve endings. Free nerve endings respond to pain and temperature. Ruffini's corpuscles respond to continuous pressure. Pacinian corpuscles respond to vibration and rapidly changing pressure. Krause's end bulbs are a receptor for fine touch which are located in mucous membranes and the tongue. Meissner's corpuscles are also a receptor for fine touch but they are located in the dermis.

6.

What is the correct term for cuticle?

- a. Lunula
- b. Eponychium
- c. Matrix
- d. Nail bed
- e. Root

Answer: b

The lunula is the half moon shaped white area on a nail. The anatomical term for the cuticle is the eponychium. The matrix is the region of the nails where there are dividing cells and nail growth. The nail plate rests on the nail bed. The nail root is the proximal portion of the nail that is underneath skin.

7.

What color is keratin with Masson's trichrome stain?

- a. Red
- b. Pink
- c. Green
- d. Black
- e. Yellow

Answer: a

A trichrome stain is a mixture of three dyes. Collagen fibers stain green with Masson's trichrome stain. Muscle and keratin will be red with Masson's trichrome stain. Cytoplasm will be pink. Nuclei will be black.

8.

What type of tissue makes up the dermis of the skin?

- a. Mucous connective tissue
- b. Mesenchyme
- c. Loose irregular connective tissue
- d. Dense irregular connective tissue
- e. Dense regular connective tissue

Answer: d

Mesenchyme is embryonic connective tissue. It is an undifferentiated tissue found in the embryo. Mucous connective tissue is a type of embryonic connective tissue; it is a subset of mesenchyme. Wharton's jelly is mucous connective tissue. Loose irregular connective tissue is areolar tissue. Dense irregular connective tissue is seen in the dermis. Dense regular connective tissue comprises tendons and ligaments.

9.

What are the pressure receptors in skin called?

- a. Psammoma bodies
- b. Corpora arenacea
- c. Hassall's corpuscles
- d. Prostatic concretions
- e. Pacinian corpuscles

Answer: e

Psammoma bodies are collections of calcium. It is derived from the Greek word "psammos", which means sand.

Corpora arenacea refers to the calcifications seen in the pineal gland. Corpora arenacea is nicknamed "brain sand".

Hassall's corpuscles are the ring like structures found in the thymus.

The spherical structures seen in some prostatic alveoli are called prostatic concretions.

Pacinian corpuscles are pressure receptors in the skin.

10.

Which sensory receptor in the skin is NOT encapsulated?

- a. Ruffini endings
- b. Free nerve endings
- c. Pacinian corpuscles
- d. Meissner's corpuscles
- e. Merkel cells

Answer: b

Ruffini endings, pacinian corpuscles, meissner's corpuscles, and merkel cells are all encapsulated sensory receptors. Free nerve endings are not encapsulated.

Sebaceous Gland

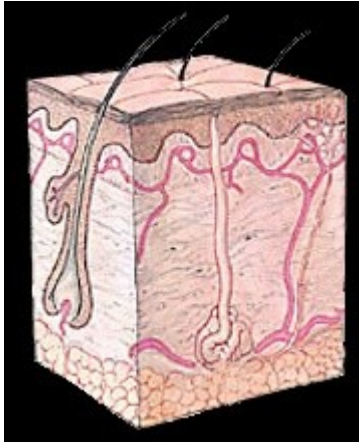
This is a histology slide of a sebaceous gland. It appears as a foamy structure lying in the dermis. There are polyhedral cells with central nuclei. In the periphery of the gland are the basal cells. The basal cells have more flattened nuclei. These are mitotically active cells. As the cells become filled with lipid, they move away from the periphery. The entire cell and its product will be discharged onto a hair follicle. Thus, this is a holocrine gland.

Sweat Gland

This is a histology slide of a sweat gland. There are adipose cells surrounding the gland. This section contains both the secretory portion and ducts of the gland. The ducts stain more acidophilic, that is, they stain pinker and deeper. The ducts have a slightly smaller diameter. There is lots of simple cuboidal epithelium present on this histology slide.

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- [Types of Skin](#)
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- [Histology of the Epidermis](#)

- [Histology of the Dermis](#)
- [Histology of the Subcutaneous Layer](#)
- [Receptors in the Skin](#)
- [Histology of Skin Appendages](#)



Functions of the Skin

The skin protects the body. The skin protects the body from water loss. The skin is involved in the production of vitamin D from precursors with the aid of sunlight. There are many sensory receptors in the skin: pain, pressure, fine touch. The skin is also involved in heat regulation.

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Types of Skin

Thick skin is found on the palms of the hand and the sole of the feet. Thin skin is found everywhere else.

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Layers of the Skin

The skin is composed of two layers: the epidermis and the dermis. Underneath these layers lies the hypodermis (subcutaneous tissue). The hypodermis is a layer of loose connective tissue.

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Histology of the Epidermis

The epidermis is formed by stratified squamous epithelium. Keratinization is seen in the epidermis. Keratinocytes, melanocytes, Merkel cells and Langerhans cells are all found in the epidermis. The keratinocyte is the most abundant cell in the epidermis. The melanocyte produces melanin, which is responsible for skin pigmentation. The Merkel cell is a mechanoreceptor. The Langerhans cell is a phagocyte. Langerhans cells are macrophages.

seen in the skin.

The epidermis is divided into five layers: stratum basale, stratum spinosum, stratum granulosum, stratum lucidum, and stratum corneum.

Stratum Basale

The stratum basale contains the dividing cells. This layer is also called the stratum germinativum.

Stratum Spinosum

The stratum spinosum consists of a layer several cells deep. The cells have pointy or spiny processes on them.

Stratum Granulosum

The cells in the stratum granulosum contain keratohyaline granules.

Stratum Lucidum

The stratum lucidum is present only in thick skin.

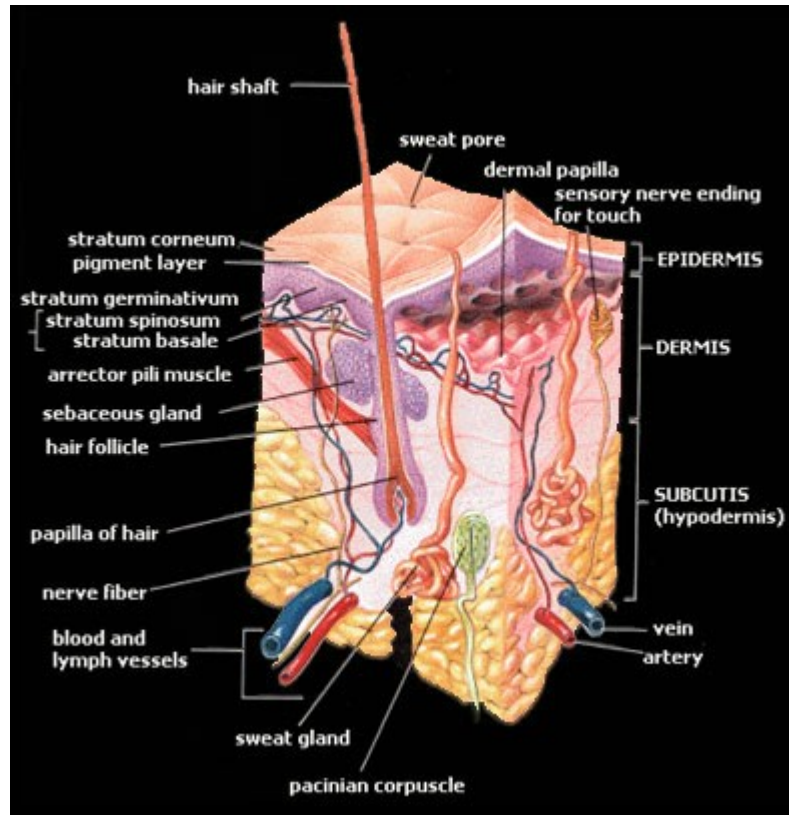
Stratum Corneum

The stratum corneum is the outermost layer. It is also called the horny layer. The cells in this layer are essentially bags of keratin. They contain no nuclei or organelles.

Histology hint from Sarah Bellham: The epithelium is classified as "squamous" based on the cells of the surface layer.

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Histology of the Dermis



Beneath the epidermis is the dermis. The dermis is composed of a papillary layer and a reticular layer. The reticular layer of the dermis is made up of dense irregular connective tissue. Fibroblasts are found in the dermis. Fibroblasts produces collagen.

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Histology of the Subcutaneous Layer

The subcutaneous tissue layer (hypodermis) is the loose connective tissue layer underneath the dermis.

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Receptors in the Skin

There are several different sensory receptors in the skin.

Ruffini endings, pacinian corpuscles, meissner's corpuscles, and merkel cells are all encapsulated sensory receptors. Free nerve endings are not encapsulated.

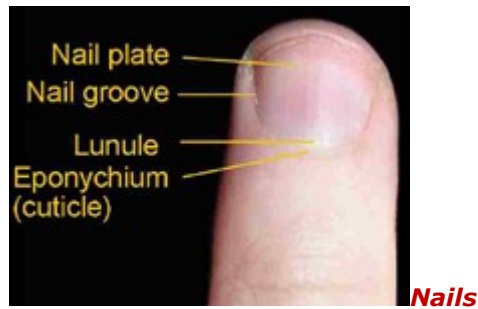
The most abundant sensory receptor are the free nerve endings.

Free nerve endings respond to pain and temperature. Ruffini's corpuscles respond to continuous pressure. Pacinian corpuscles respond to vibration and rapidly changing pressure. Krause's end bulbs are a receptor for fine touch which are located in mucous membranes and the tongue. Meissner's corpuscles are also a receptor for fine touch but they are located in the dermis. Pacinian corpuscles are pressure receptors in the skin.

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Skin Appendages

Sweat glands, hair, nails and sebaceous glands are all considered epidermal appendages.



The lunula is the half moon shaped white area on a nail. The anatomical term for the cuticle is the eponychium. The matrix is the region of the nails where there are dividing cells and nail growth. The nail plate rests on the nail bed. The nail root is the proximal portion of the nail that is underneath skin.

Ceruminous Glands

The ceruminous glands of the ear are apocrine sweat glands.

Glands of Moll

The glands of Moll in the eyelid are apocrine sweat glands.

Histology of Sweat Glands

Sweat glands are also called sudoriferous glands.

Classification of Sweat Glands

Sweat glands are divided into apocrine and eccrine. Apocrine sweat glands are found on the areola, external genitalia, axilla, and

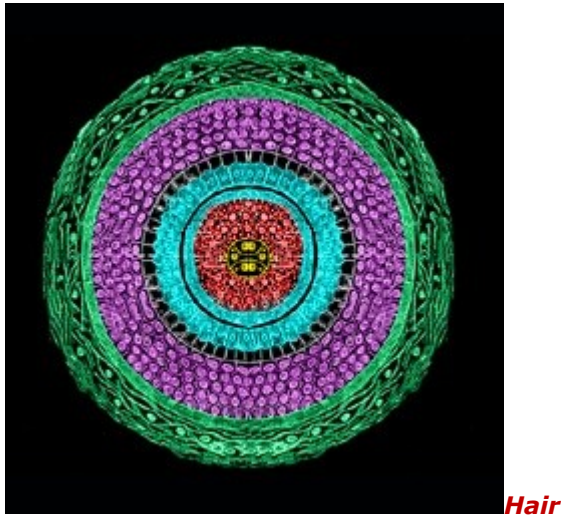
curcumanal region. Eccrine sweat glands are distributed over most of the body.

Innervation of Sweat Glands

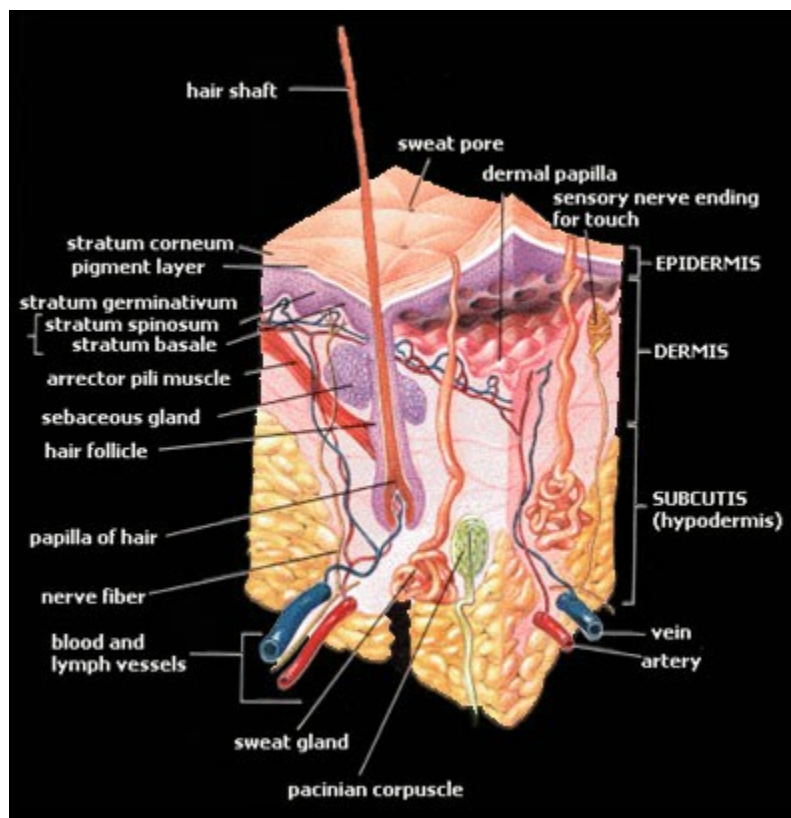
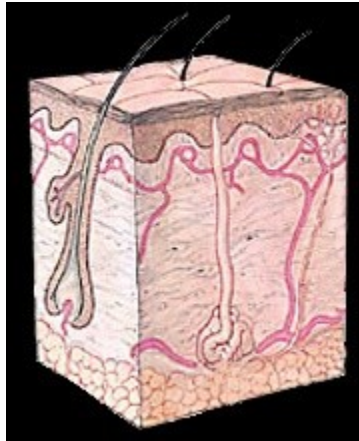
Eccrine sweat glands are innervated by the sympathetic nervous system. The neurotransmitter for the eccrine sweat glands is acetylcholine. Thus, it is cholinergic.

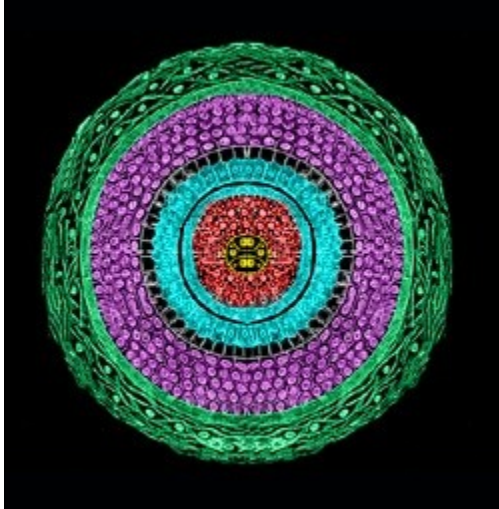
Histology hint from Sarah Bellham: For most postganglionic sympathetic neurons, the neurotransmitter is norepinephrine. Eccrine sweat glands are an exception to this generalization, as the innervation for eccrine sweat glands is cholinergic sympathetic.

Apocrine sweat glands are innervated by the sympathetic nervous system. The neurotransmitter for the apocrine sweat glands is norepinephrine. Thus, it is adrenergic.



Hair is present over most of the body. It is not found on the palms of the hand, soles of the feet, urogenital openings, and lips.





kkjj

Instructions: For each histology question, pick the one best answer. This histology test bank is also useful for the histology questions on the USMLE (USMLE step 1).

1.

Which cell is also called a septal cell?

- a. Clara cell
- b. Type I pneumocyte
- c. Type II pneumocyte
- d. Dust cell
- e. Brush cell

Answer: c

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II

pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

2.

Which cell is a respiratory macrophage?

- a. Kupffer cells
- b. Histiocyte
- c. Dust cell
- d. Langerhans cell
- e. Microglia

Answer: c

Macrophages are mononuclear phagocytes. Many tissues have resident (fixed) macrophages. Fixed macrophages are given a unique name, depending on the tissue that they are located in. Kupffer cells are the hepatic macrophages. Histiocytes are macrophages seen in connective tissue. Dust cells are alveolar macrophages found in the respiratory tract. Langerhans cells are macrophages seen in the skin. Microglia are the central nervous system macrophages.

3.

Which cell is a type of neuron?

- a. Basal cells
- b. Brush cells
- c. Olfactory cells
- d. Sustentacular cells
- e. All of the above

Answer: c

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

4.

What cell types are found in the respiratory mucosa?

- a. Ciliated cells
- b. Goblet cells
- c. Basal cells
- d. Brush cells
- e. All of the above

Answer: e

In man, respiratory mucosa is composed of ciliated pseudostratified columnar epithelium with goblet cells. Respiratory mucosa contains several cell types: ciliated cells, goblet cells, basal cells, and brush cells. Respiratory mucosa is located lining the respiratory segment of the nasal cavity. It lines the conchae and the paranasal sinuses.

5.

What type of tissue makes up the epiglottis?

- a. Compact bone
- b. Spongy bone
- c. Hyaline cartilage
- d. Fibrocartilage
- e. Elastic cartilage

Answer: e

The epiglottis is part of the larynx. It is composed of elastic cartilage.

6.

What is the smallest airspace of the respiratory tree?

- a. Alveolar duct
- b. Alveoli
- c. Alveolar sac
- d. Respiratory bronchiole
- e. Terminal bronchiole

Answer: b

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.

7.

What type of epithelium lines the trachea?

- a. Simple squamous epithelium
- b. Simple cuboidal epithelium
- c. Simple columnar epithelium
- d. Stratified squamous epithelium
- e. Pseudostratified epithelium

Answer: e

The trachea is lined by pseudostratified squamous epithelium. Epithelium lines body cavities and surfaces. Pseudostratified squamous epithelium is "pseudostratified" because it is only one cell layer thick, yet it appears to be stratified. In reality, every cell touches the basement membrane.

Note from Sarah Bellham: The prefix "pseudo" is of Greek origin and it means false or counterfeit. For example: pseudonym, pseudo-science or pseudostratified.

8.

Which structure is part of the conducting portion of the airway?

- a. Bronchi
- b. Alveolar ducts
- c. Alveoli
- d. Alveolar sacs
- e. Respiratory bronchioles

Answer: a

The conducting portion of the airway is where air is moved, warmed and moistened. The nasal cavities, pharynx, larynx, trachea and bronchi are all part of the conducting portion of the airway.

The respiratory portion of the airway is where gas exchange occurs. The respiratory bronchioles, alveolar ducts, alveolar sacs and alveoli are all part of the respiratory portion.

9.

What type of epithelium lines the vestibule?

- a. Simple squamous epithelium
- b. Simple columnar epithelium
- c. Stratified squamous epithelium
- d. Ciliated pseudostratified epithelium with goblet cells
- e. Transitional epithelium

Answer: c

The vestibule is lined by stratified squamous epithelium.

10.

Which cartilage of the larynx is made of hyaline cartilage?

- a. Thyroid cartilage
- b. Cricoid cartilage
- c. Arytenoid cartilage
- d. Corniculate cartilage
- e. All of the above

Answer: e

The larynx is composed of several cartilages. The thyroid cartilage, cricoid cartilage, arytenoid cartilages, corniculate cartilages and cuneiform cartilages are all composed of hyaline cartilage. The epiglottis is elastic cartilage. There is no fibrocartilage in the larynx.

- 1.**
What part of the respiratory tree is the functional unit where gas exchange occurs?
- a. Alveolar duct***
 - b. Alveoli***
 - c. Alveolar sac***
 - d. Respiratory bronchiole***
 - e. Terminal bronchiole***

Answer: b

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is analogous to a thoroughfare with many cul-de-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.

- 2.**
Which cell type is involved in general sensation of the olfactory mucosa?
- a. Basal cells***
 - b. Brush cells***
 - c. Olfactory cells***
 - d. Sustentacular cells***
 - e. All of the above***

Answer: b

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

3.

What type of cells are found in the olfactory mucosa?

- a. Basal cells**
- b. Brush cells**
- c. Olfactory cells**
- d. Sustentacular cells**
- e. All of the above**

Answer: e

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

4.

What type of tissue makes up the rings of the trachea?

- a. Compact bone**
- b. Spongy bone**
- c. Hyaline cartilage**
- d. Fibrocartilage**
- e. Elastic cartilage**

Answer: c

The rings of the trachea are composed of hyaline cartilage.

5.

Which cell is a squamous pulmonary epithelial cell?

- a. Clara cell**
- b. Type I pneumocyte**
- c. Type II pneumocyte**
- d. Dust cell**
- e. Brush cell**

Answer: b

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

6.

Which cell is also called an alveolar phagocyte?

- a. Clara cell***
- b. Type I pneumocyte***
- c. Type II pneumocyte***
- d. Dust cell***
- e. Brush cell***

Answer: d

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

7.

Which is structure is NOT part of the conducting portion of the airway?

- a. Bronchi***
- b. Larynx***
- c. Trachea***
- d. Larynx***
- e. Respiratory bronchioles***

Answer: e

The conducting portion of the airway is where air is moved, warmed and moistened. The nasal cavities, pharynx, larynx, trachea and bronchi are all part of the conducting portion of the airway.

The respiratory portion of the airway is where gas exchange occurs. The respiratory bronchioles, alveolar ducts, alveolar sacs and alveoli are all part of the respiratory portion.

8.

What type of tissue lines the pharynx?

- a. Simple squamous epithelium***
- b. Simple columnar epithelium***
- c. Stratified squamous epithelium***
- d. Ciliated pseudostratified epithelium with goblet cells***
- e. Both c and d***

Answer: e

The pharynx is lined by both stratified squamous epithelium and ciliated pseudostratified epithelium with goblet cells. Different regions are lined by a different type of epithelium. Regions of the pharynx that are likely to be roughened up by food are lined by stratified squamous epithelium. Other regions of the pharynx are lined by ciliated pseudostratified epithelium with goblet cells.

9.

What is an olfactory cell?

- a. Unipolar neuron**
- b. Bipolar neuron**
- c. Multipolar neuron**
- d. Supporting cell**
- e. None of the above**

Answer: b

An olfactory cell is a bipolar neuron.

10.

Which type of cartilage is found in the larynx?

- a. Hyaline cartilage**
- b. Elastic cartilage**
- c. Fibrocartilage**
- d. Both a and b**
- e. All of the above**

Answer: d

The larynx is composed of several cartilages. The thyroid cartilage, cricoid cartilage, arytenoid cartilages, corniculate cartilages and cuneiform cartilages are all composed of hyaline cartilage. The epiglottis is elastic cartilage. There is no fibrocartilage in the larynx.

1.

What part of the respiratory tree is the functional unit where gas exchange occurs?

- a. Alveolar duct**
- b. Alveoli**
- c. Alveolar sac**
- d. Respiratory bronchiole**
- e. Terminal bronchiole**

Answer: b

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.

2.

Which cell type is involved in general sensation of the olfactory mucosa?

- a. Basal cells**
- b. Brush cells**
- c. Olfactory cells**
- d. Sustentacular cells**
- e. All of the above**

Answer: b

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

3.

What type of cells are found in the olfactory mucosa?

- a. Basal cells**
- b. Brush cells**
- c. Olfactory cells**
- d. Sustentacular cells**
- e. All of the above**

Answer: e

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

4.

What type of tissue makes up the rings of the trachea?

- a. Compact bone***
- b. Spongy bone***
- c. Hyaline cartilage***
- d. Fibrocartilage***
- e. Elastic cartilage***

Answer: c

The rings of the trachea are composed of hyaline cartilage.

5.

Which cell is a squamous pulmonary epithelial cell?

- a. Clara cell***
- b. Type I pneumocyte***
- c. Type II pneumocyte***
- d. Dust cell***
- e. Brush cell***

Answer: b

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

6.

Which cell is also called an alveolar phagocyte?

- a. Clara cell***
- b. Type I pneumocyte***

- c. Type II pneumocyte**
- d. Dust cell**
- e. Brush cell**

Answer: d

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

7.

Which structure is NOT part of the conducting portion of the airway?

- a. Bronchi**
- b. Larynx**
- c. Trachea**
- d. Larynx**
- e. Respiratory bronchioles**

Answer: e

The conducting portion of the airway is where air is moved, warmed and moistened. The nasal cavities, pharynx, larynx, trachea and bronchi are all part of the conducting portion of the airway.

The respiratory portion of the airway is where gas exchange occurs. The respiratory bronchioles, alveolar ducts, alveolar sacs and alveoli are all part of the respiratory portion.

8.

What type of tissue lines the pharynx?

- a. Simple squamous epithelium**
- b. Simple columnar epithelium**
- c. Stratified squamous epithelium**
- d. Ciliated pseudostratified epithelium with goblet cells**
- e. Both c and d**

Answer: e

The pharynx is lined by both stratified squamous epithelium and ciliated pseudostratified epithelium with goblet cells. Different regions are lined by a different type of epithelium.

Regions of the pharynx that are likely to be roughened up by food are lined by stratified squamous epithelium. Other regions of the pharynx are lined by ciliated pseudostratified epithelium with goblet cells.

- 9.**
What is an olfactory cell?
a. Unipolar neuron
b. Bipolar neuron
c. Multipolar neuron
d. Supporting cell
e. None of the above

Answer: b

An olfactory cell is a bipolar neuron.

- 10.**
Which type of cartilage is found in the larynx?
a. Hyaline cartilage
b. Elastic cartilage
c. Fibrocartilage
d. Both a and b
e. All of the above

Answer: d

The larynx is composed of several cartilages. The thyroid cartilage, cricoid cartilage, arytenoid cartilages, corniculate cartilages and cuneiform cartilages are all composed of hyaline cartilage. The epiglottis is elastic cartilage. There is no fibrocartilage in the larynx.

Instructions: For each histology question, pick the one best answer. This histology test bank is also useful for the histology questions on the USMLE (USMLE step 1).

- 1.**
What structure is similar to a "bunch of grapes?"
a. Alveolar duct
b. Alveoli
c. Alveolar sac
d. Respiratory bronchiole
e. Terminal bronchiole

Answer: c

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.

2.

Which cell is found in large numbers in the terminal bronchioles?

- a. Clara cell*
- b. Type I pneumocyte*
- c. Type II pneumocyte*
- d. Dust cell*
- e. Brush cell*

Answer: a

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

3.

Which cell type is located at the basal lamina of the olfactory mucosa?

- a. Basal cells*
- b. Brush cells*
- c. Olfactory cells*
- d. Sustentacular cells*
- e. All of the above*

Answer: a

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

4.

Which cell type is most numerous in olfactory mucosa?

- a. Basal cells
- b. Brush cells
- c. Olfactory cells
- d. Sustentacular cells
- e. None of the above

Answer: d

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

5.

What type of epithelium is found in the respiratory mucosa of man?

- a. Non-ciliated pseudostratified columnar epithelium with goblet cells
- b. Ciliated pseudostratified columnar epithelium with goblet cells
- c. Simple columnar epithelium
- d. Stratified squamous epithelium
- e. Transitional epithelium

Answer: b

In man, respiratory mucosa is composed of ciliated pseudostratified columnar epithelium with goblet cells. Respiratory mucosa contains several cell types: ciliated cells, goblet cells, basal cells, and brush cells. Respiratory mucosa is located lining the respiratory segment of the nasal cavity. It lines the conchae and the paranasal sinuses.

6.

What type of tissue makes up the "Adam's apple"?

- a. Compact bone
- b. Spongy bone
- c. Hyaline cartilage
- d. Fibrocartilage
- e. Elastic cartilage

Answer: e

The "Adam's apple" is a nickname for part of the larynx formed by the thyroid cartilage. The thyroid cartilage is composed of hyaline cartilage.

7.

What type of tissue forms the alveoli in the lung?

- a. Simple squamous epithelium
- b. Simple cuboidal epithelium
- c. Simple columnar epithelium
- d. Stratified squamous epithelium
- e. Pseudostratified epithelium

Answer: a

The alveoli are formed by simple squamous epithelium. Epithelium lines body cavities and surfaces. Simple squamous epithelium is "simple" because it is one cell thick. "Squamous" refers to the fact that the cells are flat.

8.

What type of tissue lines the paranasal sinuses?

- a. Simple squamous epithelium
- b. Simple columnar epithelium
- c. Stratified squamous epithelium
- d. Ciliated pseudostratified epithelium with goblet cells
- e. Transitional epithelium

Answer: d

The paranasal sinuses are lined by ciliated pseudostratified epithelium with goblet cells.

9.

In which structure does gas exchange NOT occur?

- a. Alveolar duct
- b. Alveoli
- c. Alveolar sac
- d. Respiratory bronchiole
- e. Terminal bronchiole

Answer: e

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.

10.

Which cell secretes surfactant?

- a. Clara cell**
- b. Type I pneumocyte**
- c. Type II pneumocyte**
- d. Dust cell**
- e. Brush cell**

Answer: c

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

Instructions: For each histology question, pick the one best answer. This histology test bank is also useful for the histology questions on the USMLE (USMLE step 1).

1.

Which cartilage of the larynx is made of elastic cartilage?

- a. Thyroid cartilage**
- b. Cricoid cartilage**
- c. Arytenoid cartilage**
- d. Epiglottis**
- e. Corniculate cartilage**

Answer: d

The larynx is composed of several cartilages. The thyroid cartilage, cricoid cartilage, arytenoid cartilages, corniculate cartilages and cuneiform cartilages are all composed of hyaline cartilage. The epiglottis is elastic cartilage. There is no fibrocartilage in the larynx.

2.

Which cell is rarely found in the alveolus?

- a. Clara cell**
- b. Type I pneumocyte**
- c. Type II pneumocyte**
- d. Dust cell**
- e. Brush cell**

Answer: e

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

3.

What are the alveolar pores called?

- a. Pores of Luschka**
- b. Pores of Descemet**
- c. Pores of Mall**
- d. Pores of Kohn**
- e. Pores of Disse**

Answer: d

The alveolar pores are the pores of Kohn. These are openings between adjacent alveoli.

4.

What is the first portion of the respiratory tree where gas exchange can occur?

- a. Alveolar duct**
- b. Alveoli**
- c. Alveolar sac**
- d. Respiratory bronchiole**
- e. Terminal bronchiole**

Answer: d

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much

like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.

The conducting portion of the airway is where air is moved, warmed and moistened. The nasal cavities, pharynx, larynx, trachea and bronchi are all part of the conducting portion of the airway.

The respiratory portion of the airway is where gas exchange occurs. The respiratory bronchioles, alveolar ducts, alveolar sacs and alveoli are all part of the respiratory portion.

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Histology of the Pharynx

The pharynx is lined by both stratified squamous epithelium and ciliated pseudostratified epithelium with goblet cells. Different regions are lined by a different type of epithelium. Regions of the pharynx that are likely to be roughened up by food are lined by stratified squamous epithelium. Other regions of the pharynx are lined by ciliated pseudostratified epithelium with goblet cells.

The vestibule is lined by stratified squamous epithelium.

Histology of the Nasal Cavity

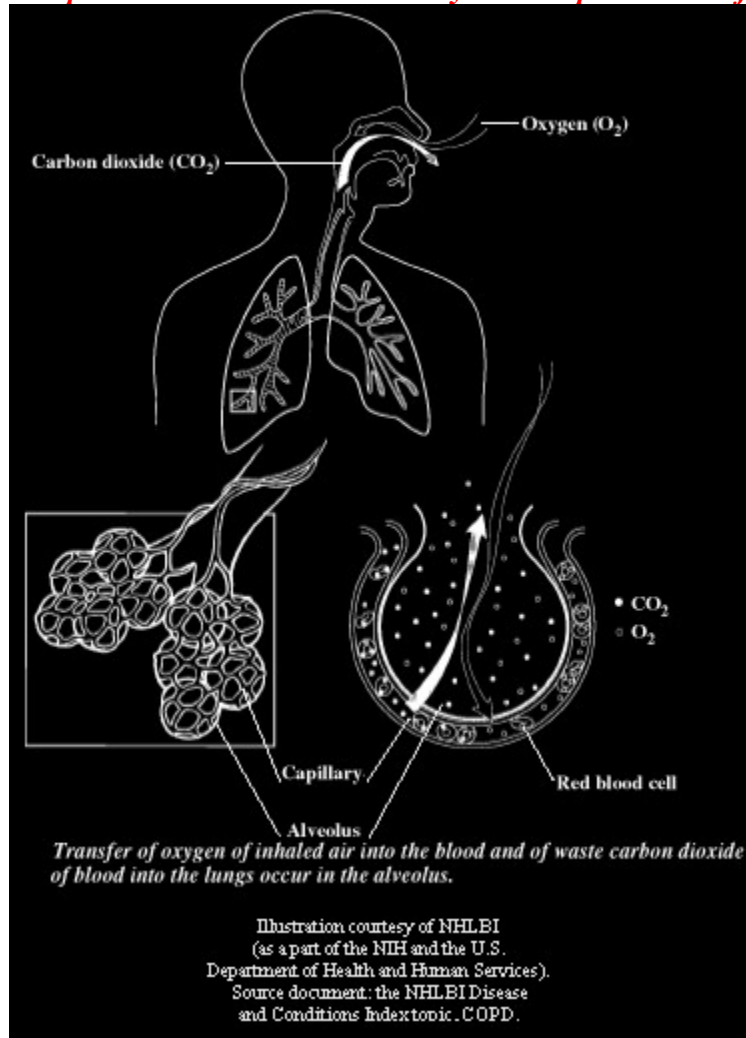
Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

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Histology of the Paranasal Sinuses

The paranasal sinuses are lined by ciliated pseudostratified epithelium with goblet cells.



In man, respiratory mucosa is composed of ciliated pseudostratified columnar epithelium with goblet cells. Respiratory mucosa contains several cell types: ciliated cells, goblet cells, basal cells, and brush cells. Respiratory mucosa is located lining the respiratory segment of the nasal cavity. It lines the conchae and the paranasal sinuses.

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Histology of the Larynx

The epiglottis is part of the larynx. It is composed of elastic cartilage.

The "Adam's apple" is a nickname for part of the larynx formed by the thyroid cartilage. The thyroid cartilage is composed of hyaline cartilage.

The larynx is composed of several cartilages. The thyroid cartilage, cricoid cartilage, arytenoid cartilages, corniculate cartilages and cuneiform cartilages are all composed of hyaline cartilage. The epiglottis is elastic cartilage. There is no fibrocartilage in the larynx.

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Histology of the Trachea

The trachea branches into two bronchi which branch into smaller bronchi. The rings of the trachea are composed of hyaline cartilage.

The trachea is lined by pseudostratified squamous epithelium. Epithelium lines body cavities and surfaces. Pseudostratified squamous epithelium is "pseudostratified" because it is only one cell layer thick, yet it appears to be stratified. In reality, every cell touches the basement membrane.

Histology hint from Sarah Bellham: The prefix "pseudo" is of Greek origin and it means false or counterfeit. For example: pseudonym, pseudo-science or pseudostratified.

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Histology of the Respiratory Tract

The respiratory tract is made of branching structures, much like the branches of a tree.

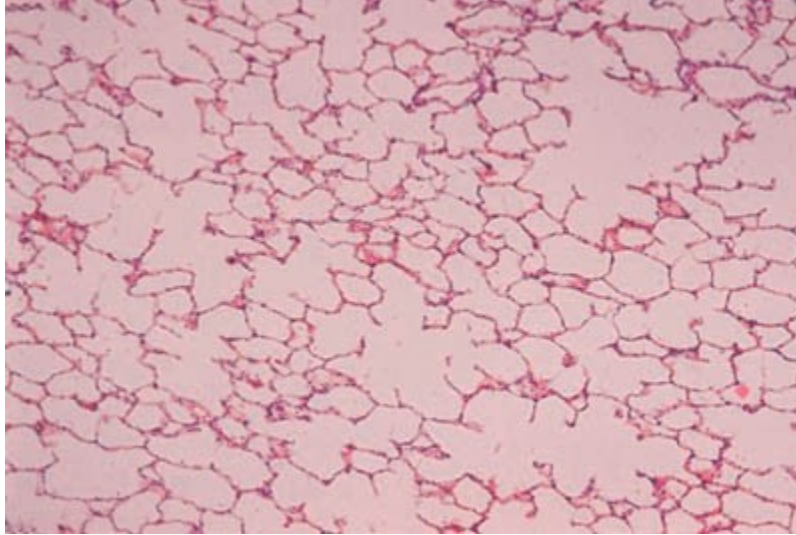
Histology of the Bronchi

The bronchi ultimately branch into smaller bronchioles.

Histology of the Bronchioles and Alveoli

Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur.

The alveoli duct is analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.



Histology slide of the lung

Histology slide courtesy of Education Interactive, Histology Photo CD

The alveoli are formed by simple squamous epithelium. Epithelium lines body cavities and surfaces. Simple squamous epithelium is "simple" because it is one cell thick. "Squamous" refers to the fact that the cells are flat.

The Clara cell is found in the terminal bronchioles.

The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte.

Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

The alveolar pores are the pores of Kohn. These are openings between adjacent alveoli.

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Macrophages are mononuclear phagocytes. Many tissues have resident (fixed) macrophages. Fixed macrophages are given a unique name, depending on the tissue that they are located in. Dust cells are alveolar macrophages found in the respiratory tract.

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Sources:

[Histology: A Text and Atlas](#)

Michael H. Ross/Edward J. Reith

<http://www.training.seer.cancer.gov>; funded by the U.S. National Cancer Institute's Surveillance, Epidemiology and End Results (SEER) Program, via contract number N01-CN-67006, with Emory University, Atlanta SEER Cancer Registry, Atlanta, Georgia, U.S.A.

QUESTION SET II. Skin. Multiple choice. [reference: <http://www.siumed.edu/~dking2/intro/skin.htm>]

Point to an answer. **Green color and bold indicates "CORRECT."** **Red color and italics indicates "Wrong answer."** (NOTE: In cases where all of the responses are correct, only "*all of the above*" will be indicated as correct.)

1. The epithelial layer of skin is:
 - a. [epidermis.](#)
 - b. [dermis.](#)
 - c. [hypodermis.](#)
 - d. [lamina propria.](#)
 - e. [mucosa.](#)

-
2. The fibrous connective tissue layer of skin is:
 - a. [epidermis.](#)
 - b. [dermis.](#)
 - c. [hypodermis.](#)
 - d. [lamina propria.](#)
 - e. [mucosa.](#)

-
3. The loose, adipose connective tissue layer of skin is:
 - a. [epidermis.](#)
 - b. [dermis.](#)
 - c. [hypodermis.](#)
 - d. [lamina propria.](#)
 - e. [mucosa.](#)

-
4. Which connective tissue layer lies closest to the epidermis:
 - a. [reticular layer of dermis.](#)
 - b. [papillary layer of dermis.](#)
 - c. [hypodermis.](#)

-
5. The projections of dermis upward, indenting the bottom of the epidermis, are called:
 - a. [arrector pili.](#)
 - b. [epidermal appendages.](#)

- c. [Meissner's corpuscles.](#)
- d. [dermal papillae.](#)
- e. [sweat ducts.](#)
-
-
- 6. Which type of encapsulated nerve ending is located in dermal papillae?
 - a. [Pacinian corpuscle.](#)
 - b. [Meissner's corpuscle.](#)
 - c. [free nerve ending.](#)
 - d. [annulospiral ending.](#)
 - e. [Merkel's corpuscle.](#)-
-
- 7. Which type of encapsulated nerve ending consists of many concentric layers (like an onion) located deep in the dermis or hypodermis?
 - a. [Pacinian corpuscle.](#)
 - b. [Meissner's corpuscle.](#)
 - c. [free nerve ending.](#)
 - d. [annulospiral ending.](#)
 - e. [Merkel's corpuscle.](#)-
-
- 8. Which of the following layers lies nearest the outer surface of the epidermis?
 - a. [stratum basale or stratum germinativum.](#)
 - b. [stratum spinosum or stratum Malpighii.](#)
 - c. [stratum granulosum.](#)
 - d. [stratum lucidum.](#)
 - e. [stratum corneum.](#)-
-
- 9. Which of the following layers lies nearest the dermis?
 - a. [stratum basale or stratum germinativum.](#)
 - b. [stratum spinosum or stratum Malpighii.](#)
 - c. [stratum granulosum.](#)
 - d. [stratum lucidum.](#)
 - e. [stratum corneum.](#)-
-
- 10. The basement membrane lies between:
 - a. [stratum basale and stratum spinosum.](#)
 - b. [stratum spinosum and stratum granulosum.](#)
 - c. [stratum granulosum and stratum corneum.](#)
 - d. [epidermis and dermis.](#)
 - e. [papillary and reticular layers of dermis.](#)-
-

11. In which of the following layers does cell division normally occur?

- a. [stratum basale or stratum germinativum.](#)
- b. [stratum spinosum or stratum Malpighii.](#)
- c. [stratum granulosum.](#)
- d. [stratum lucidum.](#)
- e. [stratum corneum.](#)

-

12. In which of the following layers do basophilic granules provide a distinct indication of the formation of a keratinized epithelium?

- a. [stratum basale or stratum germinativum.](#)
- b. [stratum spinosum or stratum Malpighii.](#)
- c. [stratum granulosum.](#)
- d. [stratum lucidum.](#)
- e. [stratum corneum.](#)

-

13. The majority of cells comprising the epidermis are called:

- a. [keratinocytes \(or "prickle cells"\).](#)
- b. [melanocytes.](#)
- c. [Langerhans cells.](#)
- d. [Merkel cells.](#)
- e. [fibroblasts.](#)

-

14. The "prickles" which characterize keratinocytes (or "prickle cells") represent the location of:

- a. [ribosomes.](#)
- b. [mitochondria.](#)
- c. [keratohyalin granules.](#)
- d. [lipofuscin.](#)
- e. [desmosomes.](#)

-

15. The most common cells in dermis are:

- a. [keratinocytes \(or "prickle cells"\).](#)
- b. [melanocytes.](#)
- c. [Langerhans cells.](#)
- d. [Merkel cells.](#)
- e. [fibroblasts.](#)

-

16. Which epidermal cell type is part of the immune system, with "antigen presenting" function?

- a. [keratinocytes \(or "prickle cells"\).](#)
- b. [melanocytes.](#)

- c. [Langerhans cells.](#)
- d. [Merkel cells.](#)
- e. [fibroblasts.](#)

-

-

17. Which epidermal cell type manufactures dark pigment?

- a. [keratinocytes \(or "prickle cells"\).](#)
- b. [melanocytes.](#)
- c. [Langerhans cells.](#)
- d. [Merkel cells.](#)
- e. [fibroblasts.](#)

-

-

18. Most of the melanin in the basal layer of the epidermis is located in which cell type?

- a. [keratinocytes \(or "prickle cells"\).](#)
- b. [melanocytes.](#)
- c. [Langerhans cells.](#)
- d. [Merkel cells.](#)
- e. [fibroblasts.](#)

-

-

19. Which epidermal cell type is closely associated with afferent nerve endings?

- a. [keratinocytes \(or "prickle cells"\).](#)
- b. [melanocytes.](#)
- c. [Langerhans cells.](#)
- d. [Merkel cells.](#)
- e. [fibroblasts.](#)

-

-

20. Embryonically, which epidermal cell type migrates into surface ectoderm from neural crest (and, perhaps encouraged by this migratory function, this cell type can form the most dangerously metastatic cancers)?

- a. [keratinocytes \(or "prickle cells"\).](#)
- b. [melanocytes.](#)
- c. [Langerhans cells.](#)
- d. [Merkel cells.](#)
- e. [fibroblasts.](#)

-

-

21. Which cell type is not normally found in the epidermis?

- a. [keratinocytes \(or "prickle cells"\).](#)
- b. [melanocytes.](#)
- c. [Langerhans cells.](#)
- d. [Merkel cells.](#)

- e. [fibroblasts.](#)

-

22. Which of the following consists of stratified (two-layered) cuboidal epithelium?

- a. [sebaceous gland.](#)
- b. [hair follicle.](#)
- c. [duct of sweat gland.](#)
- d. [secretory portion of sweat gland.](#)
- e. [arrector pili.](#)

-

23. Which of the following is NOT an "epidermal appendage" (i.e., NOT comprised of epithelial tissue)?

- a. [sebaceous gland.](#)
- b. [hair follicle.](#)
- c. [duct of sweat gland.](#)
- d. [secretory portion of sweat gland.](#)
- e. [arrector pili.](#)

-

24. Which of the following typically includes dead cells with pyknotic nuclei?

- a. [sebaceous gland.](#)
- b. [hair follicle.](#)
- c. [duct of sweat gland.](#)
- d. [secretory portion of sweat gland.](#)
- e. [arrector pili.](#)

-

25. Which of the following includes inconspicuous myoepithelial cells?

- a. [sebaceous gland.](#)
- b. [hair follicle.](#)
- c. [duct of sweat gland.](#)
- d. [secretory portion of sweat gland.](#)
- e. [hair shaft.](#)

-

26. Which layer is exceptionally thick in so-called "thick skin" (i.e., palmar and plantar skin)?

- a. [epidermis.](#)
- b. [dermis.](#)
- c. [hypodermis.](#)

-

27. Which structures are present in "thick skin" (i.e., palmar and plantar skin)?

- a. [sebaceous glands.](#)
- b. [hair follicles.](#)

- c. [arrector pili.](#)
- d. [sweat glands.](#)

-

28. Which cell type in the dermis is most closely associated with stimulating an inflammatory response, by secreting histamine and other chemical signals?

- a. [fibroblast.](#)
- b. [adipocyte.](#)
- c. [macrophage.](#)
- d. [lymphocyte.](#)
- e. [mast cell.](#)

-

29. Which cell type normally migrates through dermal (and other) connective tissues (coming in via blood vessels, leaving via lymphatic channels)?

- a. [fibroblast.](#)
- b. [adipocyte.](#)
- c. [macrophage.](#)
- d. [lymphocyte.](#)
- e. [mast cell.](#)

-

30. **QUESTION SET I. Concentric tissue layers of the GI tract.**

31. Point to an answer. **Green color and bold indicates "CORRECT."** *Red color and italics indicates "Wrong answer."* (NOTE: In cases where all of the responses are correct, only *"all of the above"* will be indicated as correct.)

32. **X. Sample question.**

- a. wrong answer.
- b. wrong answer.
- c. CORRECT answer.
- d. wrong answer.
- e. wrong answer.

33. **1. The term mucosa (mucous membrane) means:**

- a. a nonkeratinized stratified squamous epithelium.
- b. an epithelium which contains mucus-secreting goblet cells.
- c. a body surface onto which mucous glands secrete their product.
- d. the lining of the GI tract.
- e. the moist surface of any hollow organ that communicates with the outside world.

34. **2. Every mucosa consists of:**

- a. a simple epithelium.
- b. a stratified epithelium.
- c. a surface epithelium together with associated glands.
- d. a surface epithelial layer, with or without glands, together with an underlying layer of loose connective tissue.
- e. a surface epithelial layer, with or without glands, together with an underlying layer of loose connective tissue, and a layer of smooth muscle.

35. 3. The connective tissue layer of a mucous membrane is named:
- a. the submucosa.
 - b. subcutaneous tissue.
 - c. the papillary layer.
 - d. the reticular layer.
 - e. lamina propria.
36. 4. Lamina propria typically differs from ordinary loose connective tissue by including:
- a. abundant adipose tissue.
 - b. mucous glands.
 - c. numerous white blood cells.
 - d. serous glands.
 - e. densely interwoven collagen fibers.
37. 5. Besides loose connective tissue with numerous lymphocytes, lamina propria may contain:
- a. a network of capillaries.
 - b. some nerve and smooth muscle fibers.
 - c. lymph vessels.
 - d. lymph nodules.
 - e. all of the above.
38. 6. Lamina propria differs from dermis by having:
- a. more adipocytes.
 - b. more densely packed collagen fibers.
 - c. more lymphocytes.
 - d. fewer capillaries.
 - e. fewer macrophages and mast cells.
39. 7. The epithelium lining the *entire* GI tract is:
- a. stratified squamous, nonkeratinized, for protection.
 - b. simple squamous, for rapid diffusion across the epithelium.
 - c. simple columnar, for secretion and active absorption.
 - d. simple cuboidal, like that of most other tubular structures.
 - e. regionally specialized for regionally localized functions.
40. 8. Along most of the GI tract (except for the oral cavity and lower anal canal), the basal boundary of the mucosa is marked by:
- a. a thin and often inconspicuous layer of smooth muscle, the muscularis mucosae.
 - b. a thick layer of smooth muscle.
 - c. the myenteric plexus.
 - d. fibrous connective tissue of serosa or adventitia.
 - e. the mesothelium.
41. 9. The layer of tissue beneath the mucosa is called the:
- a. hypodermis.
 - b. muscularis externa.
 - c. serosa.
 - d. submucosa.
 - e. adventitia.
42. 10. Submucosa of the digestive tract consists primarily of:
- a. loose fibrous connective tissue.

- b. adipose connective tissue.
 - c. dense fibrous connective tissue.
 - d. connective tissue with interwoven bundles of smooth muscle.
 - e. lymphatic tissue.
43. 11. The submucosa differs from lamina propria by having:
- a. fewer white blood cells.
 - b. less lymphatic tissue.
 - c. larger blood vessels.
 - d. thicker and more prominent collagen fibers.
 - e. all of the above.
44. 12. Which of the following occurs within the submucosa in most regions of the GI tract:
- a. several layers of smooth muscle.
 - b. nerves and parasympathetic ganglia of the submucosal (Meissner's) plexus.
 - c. the myenteric (Auerbach's) plexus.
 - d. relatively few blood vessels.
 - e. numerous lymph nodules.
45. 13. In most regions of the GI tract (exceptions are oral cavity and stomach), smooth muscle fibers of the muscularis are arranged into:
- a. inner circular and outer longitudinal layers.
 - b. outer circular and inner longitudinal layers.
 - c. circular and longitudinal layers whose relative position varies from region to region.
 - d. an interwoven meshwork of circular and longitudinal muscle fibers.
 - e. bundles aligned along three mutually perpendicular directions.
46. 14. The myenteric (Auerbach's) plexus is located:
- a. just beneath the epithelium of the mucosa, in lamina propria.
 - b. within lamina propria, adjacent to the muscularis mucosae.
 - c. in submucosa.
 - d. between the layers of smooth muscle in the muscularis.
 - e. in serosa and/or adventitia.
47. 15. The myenteric (Auerbach's) plexus does *not* contain:
- a. postganglionic axons from sympathetic ganglia.
 - b. postganglionic axons from parasympathetic ganglia.
 - c. nerve cell bodies in sympathetic ganglia.
 - d. nerve cell bodies in parasympathetic ganglia.
 - e. preganglionic parasympathetic axons.
48. 16. For an organ lying within a body cavity, the tissue which forms the deepest layer (i.e., farthest from the inner core or lumen of the organ) is called the:
- a. mucosa.
 - b. serosa.
 - c. stroma.
 - d. parenchyma.
 - e. adventitia.
49. 17. For an organ embedded within the body wall, the tissue which forms the deepest layer (i.e., farthest from the inner core or lumen of the organ) is called the:
- a. mucosa.

- b. serosa.
 - c. stroma.
 - d. parenchyma.
 - e. adventitia.
50. 18. Adventitia consists of:
- a. surface epithelium.
 - b. adipose connective tissue.
 - c. ordinary connective tissue.
 - d. lymphatic connective tissue.
 - e. muscle tissue.
51. 19. The serosal surface exposed to a body cavity consists of:
- a. endothelium.
 - b. stroma.
 - c. mesentery.
 - d. mesothelium.
 - e. parenchyma.
52. 20. Mesothelium consists of:
- a. stratified squamous epithelium.
 - b. simple squamous epithelium.
 - c. simple cuboidal epithelium.
 - d. simple columnar epithelium.
 - e. stratified cuboidal epithelium.
53. 21. The mesentery and lining of the peritoneal cavity, like the outermost layer of the digestive tract, consist of:
- a. a mucous membrane, or mucosa.
 - b. a serous membrane, or serosa.
 - c. a submucous membrane, or submucosa.
 - d. an endothelial membrane, or endothelium.
 - e. a plasma membrane.
- 54.

QUESTION SET II. Special features of the GI tract.

Point to an answer. **Green color and bold indicates "CORRECT."** *Red color and italics indicates "Wrong answer."* (NOTE: In cases where all of the responses are correct, only "*all of the above*" will be indicated as correct.)

X. Sample question.

- a. wrong answer.
- b. wrong answer.
- c. CORRECT answer.
- d. wrong answer.
- e. wrong answer.

22. Tonsils consist of:

- a. ordinary connective tissue with many vascular sinusoids.

- b. adipose connective tissue.
- c. serous glandular tissue with vascular sinusoids.
- d. mucous glandular tissue and stroma.
- e. lymphatic connective tissue with epithelially lined crypts.

23. Tonsils occur in each of the following sites EXCEPT:

- a. in the soft palate.
- b. in the pharynx.
- c. in the posterior tongue.
- d. in the esophagus.

24. The crypts of lingual and palatine tonsils are lined by:

- a. simple squamous epithelium.
- b. stratified squamous, keratinized epithelium.
- c. stratified squamous, nonkeratinized epithelium.
- d. simple cuboidal epithelium.
- e. ciliated pseudostratified columnar epithelium.

25. The crypts of pharyngeal tonsils are lined by:

- a. simple squamous epithelium.
- b. stratified squamous, keratinized epithelium.
- c. stratified squamous, nonkeratinized epithelium.
- d. simple cuboidal epithelium.
- e. ciliated pseudostratified columnar epithelium.

26. The epithelium lining tonsillar crypts may be obscured by numerous cells migrating through it. These cells are:

- a. neutrophils.
- b. lymphocytes.
- c. mast cells.
- d. macrophages.
- e. fibroblasts.

27. Lymphocytes proliferate in relatively permanent sites in connective tissue which contain germinal centers surrounded by dense concentrations of incoming and outgoing cells. These sites are called:

- a. sinusoids.
- b. crypts.
- c. villi.
- d. lymph nodules.
- e. lymph nodes.

28. The germinal centers of lymph nodules function as sites where:

- a. previous inflammation has not yet been resolved.
- b. worn out lymphocytes are replaced.
- c. lymphocytes are secreted across mucous membranes.

- d. antigens are secreted into blood.
- e. activated lymphocytes proliferate.

29. In the digestive system, lymph nodules occur:

- a. concentrated in the tonsils.
- b. concentrated in the appendix.
- c. scattered in lamina propria throughout the GI tract.
- d. concentrated in Peyer's patches.
- e. all of the above.

30. The papillae covering most of the top surface of the human tongue (the tips of these papillae are whitish in life) are called:

- a. tonsils.
- b. fungiform papillae.
- c. filiform papillae.
- d. circumvallate papillae.
- e. dermal papillae.

31. The whitish tips of filiform papillae consist of:

- a. keratinized epithelial tissue.
- b. collagen.
- c. mucous.
- d. taste buds.
- e. goblet cells.

32. Taste buds occur most commonly:

- a. on the tips of filiform papillae.
- b. on the sides of fungiform and circumvallate papillae.
- c. in the crypts of lingual tonsils.
- d. only on the tip of the tongue.
- e. anywhere in the stratified squamous epithelium of the tongue.

33. Tubular invaginations (crypts, pits or glands) occurring within the mucosa (embedded in lamina propria), are *not* characteristic of the:

- a. esophagus
- b. stomach
- c. duodenum
- d. ileum and jejunum
- e. colon

34. Submucosal glands occur in the submucosa of the:

- a. esophagus.
- b. stomach.
- c. jejunum.
- d. ileum.
- e. colon.

35. Submucosal glands also occur in the:

- a. duodenum.
- b. ileum.
- c. jejunum.
- d. appendix.
- e. colon.

36. Folds in the mucosa of the stomach are called:

- a. rugae.
- b. plicae.
- c. villi.
- d. pits.
- e. crypts.

37. Permanent folds in the wall of the intestine, containing a core of submucosa, are called:

- a. rugae.
- b. plicae.
- c. villi.
- d. pits.
- e. crypts.

38. Microscopic finger-like projections of the intestinal mucosa are called intestinal:

- a. rugae.
- b. plicae.
- c. villi.
- d. pits.
- e. crypts.

39. Microscopic indentations in the epithelial surface of the stomach are called gastric:

- a. rugae.
- b. plicae.
- c. villi.
- d. pits.
- e. crypts.

40. Simple tubular glands within the mucosa of the small or large intestine are called intestinal:

- a. rugae.
- b. plicae.
- c. villi.
- d. pits.
- e. crypts.

41. Lymphatic vessels within intestinal villi, which transport lipids to the circulatory system, are called:

- a. capillaries.
- b. ductules.

- c. canaliculi.
- d. lacteals.
- e. crypts.

42. Taenia coli are:

- a. discrete bands of longitudinal muscle along the large intestine.
- b. crypts or mucosal glands of the large intestine.
- c. submucosal glands of the large intestine.
- d. Compact regions of lymphoid tissue in the intestine.
- e. sites of mucosal folding into the lumen of the large intestine.

QUESTION SET III. Regional differences along the GI tract.

Point to an answer. **Green color and bold indicates "CORRECT."** *Red color and italics indicates "Wrong answer."* (NOTE: In cases where all of the responses are correct, only "all of the above" will be indicated as correct.)

X. Sample question.

- a. wrong answer.
- b. wrong answer.
- c. CORRECT answer.
- d. wrong answer.
- e. wrong answer.

43. Choose the best description of the esophagus.

- a. Nonkeratinized stratified squamous epithelium; relatively unspecialized lamina propria; submucosa with scattered mucous glands.
- b. Keratinized stratified squamous epithelium; relatively unspecialized lamina propria; submucosa with scattered serous glands.
- c. Nonkeratinized stratified squamous epithelium; lamina propria with numerous lymph nodules; submucosa with scattered mucous glands.
- d. Keratinized stratified squamous epithelium; lamina propria with numerous lymph nodules; are submucosa with scattered mucous glands.
- e. Keratinized stratified squamous epithelium; relatively unspecialized lamina propria; submucosa with scattered mucous glands.

44. Choose the best description of the duodenum.

- a. Mucosa with crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.
- b. Mucosa with crypts but no villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.
- c. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa with mucous glands.
- d. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria

packed with mucous glands; submucosa with numerous lymph nodules.

e. Mucosa with crypts but no villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa with mucous glands.

45. Which of the following epithelial types is most closely associated with active absorption at a mucosal surface, as in small intestine and colon?

- a. simple squamous
- b. stratified squamous
- c. simple columnar
- d. stratified columnar
- e. stratified cuboidal

46. Which of the following epithelial types is most closely associated with active secretion at a mucosal surface, as in the stomach lining?

- a. simple squamous
- b. stratified squamous
- c. simple columnar
- d. stratified columnar
- e. stratified cuboidal

47. Choose the best description of the fundic stomach.

a. Mucosa without villi, but thick and densely packed with tubular glands containing several distinct cell types; epithelium apparently consisting of a single cell type lining surface and pits; lamina propria with numerous lymph nodules.

b. Mucosa without villi, but thick and densely packed with tubular glands containing several distinct cell types; epithelium apparently consisting of a single cell type lining surface and pits; lamina propria inconspicuous.

c. Mucosa with villi, thick and densely packed with tubular glands containing several distinct cell types; epithelium apparently consisting of a single cell type lining surface and pits; lamina propria inconspicuous.

d. Mucosa with villi, thick and densely packed with tubular glands containing several distinct cell types; epithelium with scattered goblet cells lining surface and pits; lamina propria inconspicuous.

e. Mucosa with villi, thick and densely packed with tubular glands containing several distinct cell types; epithelium with scattered goblet cells lining surface and pits; lamina propria with numerous lymph nodules.

48. Choose the best description of the jejunum.

a. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

b. Mucosa with crypts but no villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

c. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa with mucous glands.

d. Mucosa with both crypts and villi; epithelium uniform, without goblet cells; lamina propria with occasional lymph nodules; submucosa with numerous lymph nodules.

e. Mucosa with crypts but no villi; epithelium uniform, without goblet cells; lamina propria with occasional lymph nodules; submucosa with mucous glands.

49. Choose the best description of the ileum.

a. Mucosa with both crypts and villi; epithelium uniform, without goblet cells; lamina propria with occasional conspicuous clusters of lymph nodules; submucosa unspecialized.

b. Mucosa with crypts but no villi; epithelium with scattered goblet cells; lamina propria with occasional conspicuous clusters of lymph nodules; submucosa unspecialized.

c. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional conspicuous clusters of lymph nodules; submucosa with mucous glands.

d. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional conspicuous clusters of lymph nodules; submucosa unspecialized.

e. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with lymph nodules relatively uncommon; submucosa with mucous glands.

50. Choose the best description of the colon.

a. Mucosa with villi and short tubular invaginations; epithelium with many goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

b. Mucosa with short tubular invaginations but no villi; epithelium consisting entirely of goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

c. Mucosa with villi and short tubular invaginations; epithelium consisting entirely of goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

d. Mucosa thick, with many tubular glands consisting of several distinct secretory cell types; epithelium with many goblet cells; lamina propria inconspicuous; submucosa unspecialized.

e. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

51. Which of the following epithelial types is most closely associated with passive protection, without specialization for either secretion or absorption, as in the esophagus and anal canal?

a. simple squamous

b. stratified squamous

c. simple columnar

d. stratified columnar

e. stratified cuboidal

52. Which of the following regions of the GI tract is not characterized by simple columnar epithelium on the mucosal surface?

a. stomach

b. small intestine

c. appendix

d. large intestine

e. anal canal

53. Choose the best description of the appendix.

a. Mucosa with short tubular invaginations but no villi; epithelium lacking goblet cells; lamina propria packed with lymphoid tissue containing numerous germinal centers.

- b. Mucosa lacking both tubular invaginations and villi; epithelium with many goblet cells; lamina propria unspecialized, without lymphoid tissue.
- c. Mucosa lacking both tubular invaginations and villi; epithelium lacking goblet cells; lamina propria packed with lymphoid tissue containing numerous germinal centers.
- d. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria unspecialized, without lymphoid tissue.
- e. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria packed with lymphoid tissue containing numerous germinal centers.

54. Choose the best description of the gall bladder. (If you haven't learned gall bladder yet, choose the answer which does not match any region of the GI tract proper.)

- a. Mucosa with evaginations resembling villi (actually wrinkles), but no crypts; simple columnar epithelium consisting of a single absorptive cell type, without goblet cells; lamina propria unspecialized; no muscularis mucosa.
- b. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.
- c. Nonkeratinized stratified-squamous epithelium; lamina propria unspecialized; distinct muscularis mucosa; submucosa with scattered mucous glands.
- d. Mucosa without villi, but thick and densely packed with tubular glands containing several distinct cell types; epithelium apparently consisting of a single cell type lining surface and pits; lamina propria inconspicuous.
- e. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria packed with lymphoid tissue containing numerous germinal centers.

55. Which of the following regions of the GI tract is characterized by simple columnar epithelium on the mucosal surface which apparently consists of a single cell type, without goblet cells?

- a. esophagus
- b. stomach
- c. small intestine
- d. large intestine
- e. appendix

56. In which of the following regions of the GI tract does the highest proportion of goblet cells occur in the surface epithelium of the mucosa?

- a. stomach
- b. duodenum
- c. jejunum
- d. ileum
- e. large intestine

57. Which of the following regions of the GI tract is characterized by short tubular invaginations (crypts or pits) which are not lined by dividing, differentiating cells?

- a. esophagus
- b. stomach
- c. duodenum

- ☐ d. [jejunum](#)
- ☐ e. [ileum](#)

58. Which of the following regions of the GI tract is characterized by numerous short tubular invaginations but no villi?

- ☐ a. [duodenum](#)
- ☐ b. [jejunum](#)
- ☐ c. [ileum](#)
- ☐ d. [colon](#)
- ☐ e. [anal canal](#)

59. Which of the following regions of the GI tract is characterized by stratified squamous epithelium?

- ☐ a. [duodenum](#)
- ☐ b. [jejunum](#)
- ☐ c. [ileum](#)
- ☐ d. [colon](#)
- ☐ e. [anal canal](#)

60. Which of the following regions of the GI tract is characterized by villi?

- ☐ a. [esophagus](#)
- ☐ b. [stomach](#)
- ☐ c. [small intestine](#)
- ☐ d. [large intestine](#)
- ☐ e. [appendix](#)

61. Which of the following regions of the GI tract is characterized by permanent folds (plica)?

- ☐ a. [esophagus](#)
- ☐ b. [stomach](#)
- ☐ c. [small intestine](#)
- ☐ d. [large intestine](#)
- ☐ e. [appendix](#)

62. Which of the following regions of the GI tract is characterized by a thick mucosa densely packed with tubular glands containing a variety of cell types?

- ☐ a. [esophagus](#)
- ☐ b. [stomach](#)
- ☐ c. [small intestine](#)
- ☐ d. [large intestine](#)
- ☐ e. [appendix](#)

63. In which of the following regions of the GI tract does the mucosa most closely resemble that of the colon, but with more lymphoid tissue?

- ☐ a. [stomach](#)
- ☐ b. [duodenum](#)
- ☐ c. [jejunum](#)

- d. ileum
- e. appendix

64. In which region(s) of the GI tract are there tubular mucous glands in the mucosa?

- a. upper esophagus
- b. cardiac and pyloric stomach
- c. duodenum
- d. ileum
- e. appendix

65. In which region of the GI tract is the longitudinal muscle of the muscularis arranged into distinct longitudinal bundles (taenia)?

- a. stomach
- b. duodenum
- c. jejunum
- d. ileum
- e. colon

QUESTION SET IV. Glands associated with the GI tract.

Point to an answer. **Green color and bold indicates "CORRECT."** **Red color and italics indicates "Wrong answer."** (NOTE: In cases where all of the responses are correct, only "*all of the above*" will be indicated as correct.)

X. Sample question.

- a. wrong answer.
- b. wrong answer.
- c. CORRECT answer.
- d. wrong answer.
- e. wrong answer.

66. Glandular ducts are typically lined by which type of epithelium?

- a. simple squamous epithelium
- b. stratified squamous, keratinized epithelium
- c. stratified squamous, non-keratinized epithelium
- d. simple cuboidal, simple columnar, or stratified cuboidal epithelium
- e. any of the above

67. Ducts may be distinguished from secretory tissue because cells lining ducts:

- a. form a stratified epithelium.
- b. have nuclei located basally rather than apically.
- c. are usually much larger than secretory cells.
- d. lack apical vesicles and other evidence of cell specialization for synthesizing and storing

secretory product.

e. are usually more intensely stained with H&E than secretory cells.

68. The parotid gland is a:

a. serous gland, secreting a watery solution of enzymes and immunoglobulins but not mucus.

b. mucous gland, secreting mucus but not digestive enzymes.

c. mixed gland, producing both serous and mucous secretions.

69. The submandibular and sublingual glands are:

a. serous glands, secreting a watery solution of enzymes and immunoglobulins but not mucus.

b. mucous glands, secreting mucus but not digestive enzymes.

c. mixed glands, producing both serous and mucous secretions.

70. Minor salivary glands are mixed, secreting mainly mucus but also some serous product.

Minor salivary glands are found:

a. in the cheeks (buccal glands).

b. in the lips (labial glands).

c. in the soft palate (palatine glands).

d. in the tongue (lingual glands).

e. all of the above.

71. Tissue from which of the following is often poorly preserved in autopsy specimens, because it self-destructs soon after death due to post-mortem release of its stored secretory product of proteolytic enzymes?

a. submaxillary and sublingual glands

b. esophageal glands

c. duodenal (Brunner's) glands

d. parotid glands.

e. pancreas

72. The parotid gland and the pancreas are both compound acinar glands, served by a branching duct system. Morphological differences between parotid and pancreas include which of the following?

a. Islets of endocrine tissue occur in the pancreas but not the parotid.

b. Adipocytes are scattered commonly through the stroma of the parotid but not that of the pancreas.

c. Centroacinar cells (initial duct cells appearing within acini) occur in most pancreatic acini but not those of parotid.

d. all of the above.

73. Esophageal glands are located:

a. in the mucosa.

b. in the submucosa.

c. in the serosa.

- d. in the adventitia.
- e. deep to the wall of the esophagus proper.

74. Gastric glands are located:

- a. in the mucosa of the stomach.
- b. in the submucosa of the stomach.
- c. in the serosa of the stomach.
- d. deep to the stomach wall proper.
- e. more than one of the above.

75. Duodenal (Brunner's) glands are located:

- a. in the mucosa of the duodenum.
- b. in the submucosa of the duodenum.
- c. in the adventitia of the posterior duodenal wall.
- d. deep to the wall of the duodenum.
- e. more than one of the above.

76. The pancreas is located:

- a. in the mucosa of the small intestine.
- b. in the submucosa of the small intestine.
- c. in the serosa of the small intestine.
- d. deep to the wall of the small intestine proper.

77. Intestinal crypts of Lieberkuhn are located:

- a. in the mucosa of the small intestine.
- b. in the submucosa of the small intestine.
- c. in the serosa of the small intestine.
- d. deep to the wall of the small intestine proper.
- e. more than one of the above.

78. All of the following are primarily mucous-secreting glands EXCEPT:

- a. esophageal glands.
- b. cardiac glands.
- c. fundic glands.
- d. pyloric glands.
- e. duodenal (Brunner's) glands.

79. Which of the following lacks clearly differentiated ducts (i.e., consists almost entirely of secretory cells)?

- a. minor salivary glands
- b. parotid gland
- c. esophageal glands
- d. gastric glands
- e. pancreas

80. Which of the following $\diamond$ glands $\diamond$ is primarily a protected site for mitotic activity rather than a tissue specialized for secretion?

- a. parotid salivary glands.
- b. esophageal glands.
- c. gastric (fundic) glands.
- d. duodenal (Brunner $\diamond$ s) glands.
- e. intestinal crypts (of Lieberk $\diamond$ hn).

81. Which of the dimensions is about 1 $\diamond$ 2 millimeters? (Think about actual sizes as you can see them in gross lab and with your unaided eye on microscope slides, and also the sizes of familiar cells like the RBC. Remember, having a reasonable intuition about size can help you notice many significant departures from normal or expected structure.)

- a. The thickness of the stomach wall (all layers).
- b. The thickness of the gastric mucosa.
- c. The height of the stomach $\diamond$ s surface epithelial cells.
- d. The diameter of a parietal cell.
- e. The length of microvilli.

82. The length of intestinal villi is about:

- a. 10 $\diamond$ 20 millimeters.
- b. 0.5 $\diamond$ 1.0 millimeters.
- c. 20 $\diamond$ 50 microns.
- d. 1 $\diamond$ 2 microns.
- e. 50 $\diamond$ 100 nanometers.

83. The length of microvilli is about:

- a. 10 $\diamond$ 20 millimeters.
- b. 0.5 $\diamond$ 1.0 millimeters.
- c. 20 $\diamond$ 50 microns.
- d. 1 $\diamond$ 2 microns.
- e. 50 $\diamond$ 100 nanometers.

84. The principal secretory product for cells of Brunner $\diamond$ s (duodenal) glands is:

- a. acid mucus.
- b. alkaline mucus.
- c. proteolytic and other digestive enzymes.
- d. lysosomal enzymes.
- e. any one of several duodenal hormones.

QUESTION SET V. Specialized cell types of the GI tract.

Point to an answer. Green color and bold indicates "CORRECT." Red color and italics indicates "Wrong answer." (NOTE: In cases where all of the responses are correct, only "all of the above" will be indicated as correct.)

X. Sample question.

- a. wrong answer.
- b. wrong answer.
- c. CORRECT answer.
- d. wrong answer.
- e. wrong answer.

85. Which of the following is characteristic of mucus-secreting cells?

- a. polarized cytoplasm, with distinct apical and basal regions
- b. rough endoplasmic reticulum in basal cytoplasm
- c. supranuclear Golgi apparatus
- d. large, mucus-containing vesicles occupying apical cytoplasm
- e. all of the above

86. Which the following stains is used to demonstrate mucus (as well as basement membranes and glycogen)?

- a. hematoxylin
- b. eosin
- c. periodic acid Schiff (PAS)
- d. Golgi silver
- e. Mucus can only be stained immunohistochemically.

87. In H&E stained tissue sections, the mucus-containing secretory vesicles in mucous cells typically appear:

- a. red or pink.
- b. blue or purple.
- c. yellow or brown.
- d. black.
- e. clear or only weakly stained.

88. Which of the following is characteristic of serous (enzyme-secreting) cells?

- a. polarized cytoplasm, with distinct apical and basal regions
- b. extensive rough endoplasmic reticulum in basal cytoplasm
- c. supranuclear Golgi apparatus
- d. zymogen vesicles occupying apical cytoplasm
- e. all of the above

89. In H&E stained tissue sections, the basal cytoplasm in enzyme-secreting (serous) cells typically appears:

- a. red or pink.
- b. blue or purple.
- c. yellow or brown.

- d. black.
- e. clear or only weakly stained.

90. Contractile epithelial cells which, under autonomic neural control, help expel product from glandular acini and ducts are called:

- a. myoepithelial cells.
- b. mucous cells.
- c. interlobular duct cells.
- d. intralobular duct cells.
- e. enteroendocrine cells.

91. The epithelial cells lining striated intralobular ducts:

- a. are usually simple cuboidal or columnar cells.
- b. have relatively eosinophilic cytoplasm containing numerous mitochondria.
- c. have basal cell membranes extensively folded to increase the surface that contains active ion pump molecules.
- d. are specialized for concentrating secretory products in the duct lumen.
- e. All of the above.

92. Cells which secrete HCl into the stomach are called:

- a. goblet cells.
- b. chief (zymogenic) cells.
- c. parietal (oxyntic) cells.
- d. enterocytes.
- e. Paneth cells.

93. Cells which secrete pepsinogen into the stomach are called:

- a. goblet cells.
- b. chief (zymogenic) cells.
- c. parietal (oxyntic) cells.
- d. Kupffer cells.
- e. Paneth cells.

94. Enteroendocrine cells, an inconspicuous but important epithelial cell type, occur:

- a. only in the stomach (especially its pyloric region).
- b. only in the small intestine.
- c. only in the colon.
- d. only in the gall bladder.
- e. mainly in the stomach (especially its pyloric region) and in the small intestine, but also elsewhere along the GI tract (and also along respiratory passages).

95. Which of the following cell types is characterized by a large nucleus (sometimes two nuclei) surrounded by prominent eosinophilic cytoplasm containing very many mitochondria and an intracellular canaliculus?

- a. gastric chief cells
- b. gastric parietal cells

- c. enteroendocrine cells
- d. Paneth cells
- e. Kupffer cells

96. Which of the following cell types is characterized by secretory vesicles concentrated in basal cytoplasm, near the basement membrane?

- a. gastric chief cells
- b. gastric parietal cells
- c. enteroendocrine cells
- d. Paneth cells
- e. gastric surface mucous cells

97. Which of the following cell types is characterized by extensive basal rough endoplasmic reticulum, a supranuclear Golgi apparatus and apical zymogen vesicles?

- a. gastric chief cells
- b. gastric parietal cells
- c. enteroendocrine cells
- d. gastric surface mucous cells
- e. gastric mucous neck cells

98. Which of the following cells is most directly responsible for protecting the stomach mucosa from attack by acid and proteolytic enzymes?

- a. gastric chief cells
- b. gastric parietal cells
- c. enteroendocrine cells
- d. Paneth cells
- e. gastric surface mucous cells

99. Which of the following cell types is characterized by extensive basal rough endoplasmic reticulum, a supranuclear Golgi apparatus and apical zymogen vesicles?

- a. pancreatic acinar cell
- b. Brunner's gland cell
- c. intestinal goblet cell
- d. intestinal absorptive cell (enterocyte)
- e. centroacinar cell

100. The principal function for intestinal goblet cells is:

- a. secreting digestive enzyme.
- b. secreting one of several various hormones.
- c. secreting mucus.
- d. absorbing nutrients.
- e. mitotic activity for continuing replacement of surface epithelium.

101. The principal function for most surface epithelial cells in the small intestine (enterocytes) is:

- a. secreting digestive enzyme.
- b. secreting one of several various hormones.

- c. secreting mucus.
- d. absorbing nutrients.
- e. mitotic activity for continuing replacement of surface epithelium.

102. The principal function for intestinal enteroendocrine cells is:

- a. secreting digestive enzyme.
- b. secreting one of several various hormones.
- c. secreting mucus.
- d. absorbing nutrients.
- e. mitotic activity for continuing replacement of surface epithelium.

103. The principal function for cells lining the sides of intestinal crypts of Lieberkühn is:

- a. secreting digestive enzyme.
- b. secreting one of several various hormones.
- c. secreting mucus.
- d. absorbing nutrients.
- e. supplying replacement cells for epithelium of villi.

104. The principal secretory product for the Paneth cells clustered at the ends of intestinal crypts is:

- a. mucus.
- b. pepsinogen.
- c. trypsinogen.
- d. any one of several intestinal hormones.
- e. lysosomal enzymes.

105. The principal secretory product for the epithelial cells forming pancreatic islets (of Langerhans) is:

- a. mucus.
- b. pepsinogen.
- c. trypsinogen.
- d. any one of several islet hormones.
- e. lysosomal enzymes.

106. The principal secretory product for the pancreatic acinar cells is:

- a. acid mucus.
- b. alkaline mucus.
- c. proteolytic and other digestive enzymes.
- d. any one of several acinar hormones.
- e. lysosomal enzymes.

107. The principal secretory product for cells of Brunner's (duodenal) glands is:

- a. acid mucus.
- b. alkaline mucus.
- c. proteolytic and other digestive enzymes.

- d. any one of several acinar hormones.
- e. lysosomal enzymes.

108. Undifferentiated, mitotically active epithelial cells which can replace or regenerate the epithelial lining of the small intestine are located:

- a. at the tips of villi.
- b. along the entire villous surface.
- c. in the lamina propria.
- d. along the wall of intestinal crypts.
- e. at the ends (bottoms) of intestinal crypts.

109. Undifferentiated, mitotically active epithelial cells which can replace or regenerate the epithelial lining of the stomach are located:

- a. among mucous cells of the gastric surface.
- b. along the sides of gastric pits.
- c. in the lamina propria.
- d. in the neck region of gastric glands.
- e. at the deep ends (bottoms) of gastric glands.

110. Columnar cells which extend across the thickness of a stratified squamous epithelium, and which have apical microvilli and basal synaptic vesicles associated with afferent nerve endings, are characteristic of:

- a. taste buds in the tongue.
- b. lingual and palatine tonsils.
- c. the upper third of the esophagus.
- d. the lower third of the esophagus.
- e. the anal canal.

111. Which of the following connective tissue cell types is *not* typical of lamina propria?

- a. lymphocyte
- b. fibroblast
- c. macrophage
- d. mast cell
- e. adipocyte

112. The mucosal surface of the stomach is protected from digestion by:

- a. stratified squamous epithelial cells.
- b. surface mucous cells.
- c. chief cells.
- d. parietal cells.
- e. alkaline secretions from Brunner's glands.

113. Absorptive epithelial cells of the small intestine (enterocytes) are characterized by:

- a. apical brush border composed of microvilli.
- b. a glycocalyx containing various digestive enzymes.
- c. junctional complexes (terminal bars), which include tight junctions that separate apical

from lateral cell membranes.

d. cytoplasm with a supranuclear Golgi apparatus, many mitochondria, lysosomes, and extensive smooth and rough endoplasmic reticulum.

e. all of the above.

114. Goblet cells in the intestine differ from enterocytes by having:

a. a more basally located nucleus.

b. fewer microvilli.

c. large vacuoles of mucus that stain poorly with routine acidic and basic dyes (like H&E).

d. a shape which is narrow basally and broad apically.

e. all of the above.

115. The basal basophilia of serous (enzyme-secreting) cells results from the presence in basal cytoplasm of:

a. extensive rough endoplasmic reticulum.

b. extensive smooth endoplasmic reticulum.

c. extensive Golgi apparatus.

d. many mitochondria.

e. numerous folds of the plasma membrane.

116. Which of the following is *not* a distinguishing characteristic of a parasympathetic neuron cell body in a ganglion of Meissner's or Auerbach's plexus?

a. relatively large diameter

b. round, large, euchromatic nucleus

c. prominent, single nucleolus

d. cytoplasmic acidophilia

e. location in submucosa or between circular and longitudinal layers of muscularis

117. Most lymphocytes in the GI tract (and elsewhere) can be recognized by their:

a. relatively small size.

b. round heterochromatic nuclei.

c. proportionately small (inconspicuous) cytoplasmic volume, compared with the nucleus.

d. location in connective tissue, often clustered near blood vessels or epithelial tissue.

e. all of the above.

118. Plasma cells differ from other lymphocytes by having:

a. a greater volume of basophilic cytoplasm (associated with rough ER for antibody production).

b. heterochromatin arranged in a radial, spoke-wheel pattern around the nucleus.

c. cytoplasm forming a lopsided or eccentric bulge on one side of the cell.

d. all of the above.

119. With H&E stain, many columnar epithelial cells display a weakly stained region in the cytoplasm above the nucleus. This pale supranuclear region represents the location of the:

a. vesicles containing stored mucus.

b. chylomicrons.

- c. Golgi apparatus.
- d. microvilli.
- e. mitochondria.

120. The cytoplasm of gastric parietal cells stains pink with H&E because:

- a. the cytoplasm is highly acidic.
- b. the cytoplasm is highly basic.
- c. the secretory vesicles are highly acidic.
- d. the numerous mitochondria, needed for the highly energetic H⁺/Cl⁻ ion pump, are acidophilic.

121. Intensely basophilic bodies in the epithelial lining of intestinal crypts, usually located toward the apical end of the columnar cells, represent:

- a. mucus vesicles.
- b. ribosomes.
- c. chylomicrons.
- d. lipofuscin granules.
- e. condensed chromatin of mitotic figures.

QUESTION SET VI. Liver.

Point to an answer. **Green color and bold indicates "CORRECT."** *Red color and italics indicates "Wrong answer."* (NOTE: In cases where all of the responses are correct, only "all of the above" will be indicated as correct.)

X. Sample question.

- a. wrong answer.
- b. wrong answer.
- c. CORRECT answer.
- d. wrong answer.
- e. wrong answer.

122. The structural unit of liver tissue, centered around a central vein with branches of the hepatic artery, portal vein and bile duct at the corners, is called a:

- a. liver acinus.
- b. portal triad, tract or canal.
- c. classical or hepatic lobule.
- d. hepatic cord.
- e. biliary tract.

123. The hepatic lobule can also be defined as a region of hepatic tissue that is served by a single:

- a. portal vein.
- b. central vein.

- c. bile duct.
- d. hepatic artery.
- e. sinusoid.

124. The tissue region located at the corners of a classical liver lobule, containing branches of portal vein, hepatic artery and bile duct all enveloped in connective tissue, is called a:

- a. liver acinus.
- b. portal triad, tract or canal.
- c. classical or hepatic lobule.
- d. hepatic cord.
- e. biliary tract.

125. A functional unit of liver tissue served by a single terminal branch of the hepatic artery, including all the hepatic cords leading toward central veins on both sides of this arteriole, is called a:

- a. liver acinus.
- b. portal triad, tract or canal.
- c. classical or hepatic lobule.
- d. sinusoid.
- e. biliary tract.

126. Cirrhosis, or liver scarring in which functional hepatic tissue is replaced by connective tissue, can be recognized by:

- a. a greater than normal amount of connective tissue in the liver.
- b. continuous strands of connective tissue extending between portal areas.
- c. multiple bile ducts embedded in large portal areas (where the intervening hepatic parenchyma has disappeared).
- d. by using a trichrome stain, which highlights collagen.
- e. by all of the above.

127. The vessel which brings venous blood from the intestine and spleen into the liver, with branches in each portal area, is the:

- a. hepatic artery.
- b. portal vein.
- c. central vein.
- d. sinusoid.
- e. bile duct.

128. The vessel which brings oxygenated, arterial blood from the heart into the liver, with branches in each portal area, is the:

- a. hepatic artery.
- b. portal vein.
- c. central vein.
- d. sinusoid.
- e. bile duct.

129. The vessel which drains bile from the liver, with branches in each portal area, is the:

- a. [hepatic artery.](#)
- b. [portal vein.](#)
- c. [central vein.](#)
- d. [sinusoid.](#)
- e. [bile duct.](#)

130. The vessel which drains venous blood from the center of each classical lobule is the:

- a. [hepatic artery.](#)
- b. [portal vein.](#)
- c. [central vein.](#)
- d. [sinusoid.](#)
- e. [bile duct.](#)

131. A hepatic cord is:

- a. [another name for a branch of the portal vein.](#)
- b. [the tough connective tissue surrounding a portal triad.](#)
- c. [a row of adjacent hepatocytes within a liver lobule.](#)
- d. [a branch of the bile duct.](#)
- e. [a nerve within the liver.](#)

132. A vascular passageway between adjacent hepatic cords is called a:

- a. [hepatic artery.](#)
- b. [portal vein.](#)
- c. [central vein.](#)
- d. [sinusoid.](#)
- e. [bile duct.](#)

133. The term liver parenchyma refers to:

- a. [hepatocytes.](#)
- b. [portal veins and hepatic arteries.](#)
- c. [bile ducts.](#)
- d. [stored glycogen.](#)
- e. [connective tissue in the liver.](#)

134. Hepatic sinusoids are lined by:

- a. [nothing; they are free spaces between hepatic cords.](#)
- b. [stratified squamous epithelium.](#)
- c. [endothelium, like any other vascular channel.](#)
- d. [simple cuboidal epithelium.](#)
- e. [simple columnar epithelium.](#)

135. The endothelium lining hepatic sinusoids is:

- a. [continuous.](#)
- b. [fenestrated.](#)
- c. [open-ended.](#)

- d. impermeable to large molecules.
- e. nonexistent.

136. What fluid flows within hepatic sinusoids?

- a. bile
- b. lymph
- c. venous blood from intestine and spleen
- d. arterial blood from hepatic artery
- e. mixed arterial and venous blood

137. Between liver sinusoidal endothelium and the basal surfaces of hepatocytes is the:

- a. bile canaliculus.
- b. space of Disse.
- c. sinusoidal lumen.
- d. portal canal.
- e. central vein.

138. The basal surface of hepatocytes is bathed by blood plasma as it flows freely through:

- a. bile canaliculi.
- b. portal canals.
- c. the space of Disse.
- d. the central veins.
- e. the portal veins.

139. Blood cells (both RBCs and leukocytes) are retained by the sinusoidal endothelium and do not normally enter the:

- a. hepatic artery.
- b. lumen of hepatic sinusoids.
- c. space of Disse.
- d. central veins.
- e. portal veins.

140. A tiny passage between adjacent hepatocytes, lying entirely within a hepatic cord, is called a:

- a. bile canaliculus.
- b. space of Disse.
- c. sinusoidal lumen.
- d. portal canal.
- e. central vein.

141. Kupffer cells are most similar to:

- a. lymphocytes.
- b. hepatocytes.
- c. neutrophils.
- d. macrophages.
- e. bile canaliculi.

142. Kupffer cells are found most commonly within the:

- ☐ a. [gall bladder.](#)
- ☐ b. [portal canals.](#)
- ☐ c. [hepatic sinusoids.](#)
- ☐ d. [central vein.](#)
- ☐ e. [bile ducts.](#)

143. Hepatocyte function includes all of the following EXCEPT:

- ☐ a. [drug detoxification.](#)
- ☐ b. [protein synthesis.](#)
- ☐ c. [gluconeogenesis.](#)
- ☐ d. [urea formation by deamination of amino acids.](#)
- ☐ e. [breakdown of hemoglobin.](#)

144. Macrophages in the spleen and Kupffer cells in the liver perform which of the following functions?

- ☐ a. [drug detoxification](#)
- ☐ b. [antibody secretion](#)
- ☐ c. [mass storage of glycogen](#)
- ☐ d. [exocrine secretion](#)
- ☐ e. [breakdown of hemoglobin](#)

145. Hepatocytes are:

- ☐ a. [lymphoid tissue.](#)
- ☐ b. [epithelial tissue.](#)
- ☐ c. [muscle tissue.](#)
- ☐ d. [adipose tissue.](#)
- ☐ e. [nervous tissue.](#)

146. The shape of hepatocytes is best described as:

- ☐ a. [cuboidal.](#)
- ☐ b. [columnar.](#)
- ☐ c. [squamous.](#)
- ☐ d. [fusiform.](#)
- ☐ e. [stellate, with many branches.](#)

147. Hepatocytes are characterized by:

- ☐ a. [round nuclei \(occasionally paired\).](#)
- ☐ b. [plentiful rough and smooth endoplasmic reticulum.](#)
- ☐ c. [many mitochondria.](#)
- ☐ d. [numerous lysosomes.](#)
- ☐ e. [all of the above.](#)

148. In normal, H&E stained material, the cytoplasm of hepatocytes appears:

- ☐ a. [unstained.](#)
- ☐ b. [deep pink.](#)

- c. dark blue or purple.
- d. yellow, brown or black.
- e. green or sky blue.

149. Endocrine secretory products of hepatocytes include all of the following EXCEPT:

- a. glucose (from glycogen).
- b. serum albumin.
- c. prothrombin.
- d. conjugated bilirubin and bile acids.
- e. fibrinogen.

150. Exocrine secretory products of hepatocytes include which of the following?

- a. glucose (from glycogen)
- b. serum albumin
- c. prothrombin
- d. conjugated bilirubin and bile acids
- e. fibrinogen

151. Fat-storing stellate cells (Ito cells) in the space of Disse are noted for storing:

- a. glycogen.
- b. glucose.
- c. albumin.
- d. vitamin A.
- e. all of the above.

152. Hepatocytes synthesize:

- a. glycogen.
- b. cholesterol.
- c. albumin.
- d. glucose.
- e. all of the above.

153. The gall bladder is best understood as a specialized region of:

- a. a lymph node.
- b. a duct.
- c. adipose tissue.
- d. hepatocyte parenchyma.
- e. a sympathetic ganglion.

154. Like the small intestine, the gall bladder:

- a. is lined by columnar cells with a brush border of microvilli.
- b. has goblet cells in the mucosal epithelium.
- c. has crypts with Paneth cells at the bottom.
- d. has villi with lacteals in the core.
- e. has a distinct muscularis mucosa, and is further surrounded by two distinct layers of circular and longitudinal muscle.

155. Although the gall bladder lining superficially resembles the mucosa of the small intestine, the gall bladder lacks:

- a. an epithelial lining.
- b. columnar cells.
- c. microvilli.
- d. lamina propria.
- e. a muscularis mucosa, crypts, villi, and goblet cells.

Multiple choice questions.

Point to an answer. Green color and bold indicates "CORRECT." Red color and italics indicates "Wrong answer."

-
(NOTE: In cases where all of the responses are correct, only "*all of the above*" will be indicated as correct.)

1. Which epithelial cell surface specialization is used to move mucous and particles along the surface?

- a. cilia
- b. stereocilia
- c. desmosomes
- d. microvilli
- e. villi

-

2. Pseudostratified columnar, ciliated epithelium is found lining the:

- a. nasal cavities.
- b. crypts of pharyngeal tonsils.
- c. bronchi.
- d. trachea.
- e. all of the above

-

3. *The cell type in the olfactory epithelium whose nuclei are positioned most apically is the:*

- a. *olfactory receptor cell.*
- b. *goblet cell.*
- c. *sustenacular cell.*
- d. *clara cell.*
- e. *basal cell.*

-

4. *The most numerous cell type in the olfactory epithelium, whose nuclei are positioned at several levels in the lower portion of the epithelium, is the:*

- a. *olfactory receptor cell.*

- b. *goblet cell.*
- c. *sustentacular cell.*
- d. *clara cell.*
- e. *basal cell.*

-

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5. Olfactory receptor cells:

- a. have several extremely long cilia for extensive exposure of receptor surface membrand.
- b. can be replaced by cell division, when worn out or damaged.
- c. are supported by sustentacular cells.
- d. are nerve cells with axons extending through the cribriform plate into the olfactory bulb of the brain.
- e. all of the above

-

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6. Nasal mucosa includes:

- a. pseudostratified, ciliated columnar epithelium.
- b. lamina propria.
- c. seromucous glands.
- d. venous plexuses.
- e. all of the above

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7. Small sero-mucous glands may be found in the mucosa of the:

- a. nasal cavity.
- b. nasal sinuses.
- c. trachea.
- d. bronchi.
- e. all of the above

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8. *In which of the following are goblet cells most frequent?*

- a. *trachea*
- b. *bronchus*
- c. *bronchiole*
- d. *alveolus*
- e. The frequency of goblet cells in each of the above is about equal.

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9. As respiratory passages branch from trachea to bronchioles, the epithelium gradually changes from:

- a. simple squamous to ciliated, pseudostratified columnar.
- b. simple columnar to simple squamous.
- c. simple cuboidal to simple columnar.
- d. stratified squamous to simple squamous.

- e. [ciliated, pseudostratified columnar to simple cuboidal.](#)

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10. Simple squamous epithelium lines the:

- a. [nasal cavities.](#)
- b. [crypts of pharyngeal tonsils.](#)
- c. [bronchi.](#)
- d. [trachea.](#)
- e. [alveoli.](#)

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11. Alveolar type I cells are:

- a. [cuboidal cells that secrete surfactant.](#)
- b. [squamous cells involved in gas exchange.](#)
- c. [ciliated cells that move mucous.](#)
- d. [columnar cells that secrete mucous.](#)
- e. [amoeboid, phagocytic cells that clean the alveolar surface.](#)

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12. Alveolar type II cells are:

- a. [cuboidal cells that secrete surfactant.](#)
- b. [squamous cells involved in gas exchange.](#)
- c. [ciliated cells that move mucous.](#)
- d. [columnar cells that secrete mucous.](#)
- e. [amoeboid, phagocytic cells that clean the alveolar surface.](#)

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13. Alveolar dust cells (macrophages) are:

- a. [cuboidal cells that secrete surfactant.](#)
- b. [squamous cells involved in gas exchange.](#)
- c. [ciliated cells that move mucous.](#)
- d. [columnar cells that secrete mucous.](#)
- e. [amoeboid that clean the alveolar surface.](#)

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14. The entire alveolar wall or septum (separating adjacent alveoli) consists of:

- a. [surface epithelium.](#)
- b. [blood vessels.](#)
- c. [connective tissue.](#)
- d. [all of the above](#)

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15. Non-ciliated cuboidal cells in bronchiole epithelium that are thought to be involved in absorption or secretion are called:

- a. [chief cells.](#)
- b. [clara cells.](#)

- c. [goblet cells.](#)
- d. [basal cells.](#)
- e. [Type I cells.](#)

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16. The thickness of interalveolar septa separating adjacent alveoli is typically:

- a. [1 micron or less.](#)
- b. [5 to 10 microns.](#)
- c. [50 to 100 microns.](#)
- d. [500 to 1000 microns.](#)
- e. [greater than 5 mm.](#)

-

17. The thickness of the gas-exchange membrane separating alveolar air from blood in capillaries is typically:

- a. [1 micron or less.](#)
- b. [5 to 10 microns.](#)
- c. [50 to 100 microns.](#)
- d. [500 to 1000 microns.](#)
- e. [greater than 5 mm.](#)

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18. The lining of the pleural cavity is:

- a. [endothelium.](#)
- b. [mesothelium.](#)
- c. [pseudostratified columnar epithelium.](#)
- d. [type I pneumocytes.](#)
- e. [type II pneumocytes.](#)

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19. C-shaped rings that form the framework of the trachea and help keep it open are composed of:

- a. [fibro-elastic tissue.](#)
- b. [trachealis muscle.](#)
- c. [skeletal muscle.](#)
- d. [hyaline cartilage.](#)
- e. [bone.](#)

-

20. Which of the following is devoid of hyaline cartilage?

- a. [trachea](#)
- b. [bronchus](#)
- c. [larynx](#)
- d. [bronchiole](#)

- e. none of the above

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Multiple choice questions.

Point to an answer. **Green color and bold indicates "CORRECT."** *Red color and italics indicates "Wrong answer."*

(NOTE: In cases where all of the responses are correct, only "all of the above" will be indicated as correct.)

1. One lobe of the kidney:
 - a. is separated from other lobes by connective tissue septa.
 - b. is visible grossly as a rounded bulge on the kidney surface.
 - c. consists of one medullary pyramid together with the associated cortex.
 - d. drains tubular fluid into a single interlobular duct.
 - e. is all of the above.

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2. The region of the kidney containing glomeruli is called the:
 - a. cortex.
 - b. medulla.
 - c. calyx.
 - d. pelvis.
 - e. papillae.

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3. The region of the kidney containing proximal convoluted tubules is called the:
 - a. cortex.
 - b. medulla.
 - c. calyx.
 - d. pelvis.
 - e. papillae.

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4. The region of the kidney containing long loops of Henle is called the:
 - a. cortex.
 - b. medulla.
 - c. calyx.
 - d. pelvis.
 - e. papillae.

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5. The region of the kidney containing distal convoluted tubules is called the:

- a. [cortex.](#)
- b. [medulla.](#)
- c. [capsule.](#)
- d. [pelvis.](#)
- e. [hilus.](#)
-
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- 6. The region of the kidney containing larger collecting ducts is called the:
 - a. [cortex.](#)
 - b. [medulla.](#)
 - c. [calyx.](#)
 - d. [pelvis.](#)
 - e. [papilla.](#)-
-
- 7. Regions of cortical parenchyma located between medullary pyramids rather than adjacent to the surface of the kidney are called:
 - a. [medullary rays.](#)
 - b. [renal lobules.](#)
 - c. [columns of Bertin.](#)
 - d. [papillae.](#)
 - e. [calyces.](#)-
-
- 8. Regions of cortex with texture similar to that of the medulla, containing straight distal tubules and collecting ducts, are called:
 - a. [medullary rays.](#)
 - b. [renal lobules.](#)
 - c. [columns of Bertin.](#)
 - d. [papillae.](#)
 - e. [calyces.](#)-
-
- 9. A region of cortex surrounding and drained by a single bundle of collecting ducts is called a:
 - a. [medullary ray.](#)
 - b. [renal lobule.](#)
 - c. [column of Bertin.](#)
 - d. [papilla.](#)
 - e. [calyx.](#)-
-
- 10. The end of each medullary pyramid through which collecting ducts open into a calyx is called a:
 - a. [medullary ray.](#)
 - b. [renal lobule.](#)

- c. [column of Bertin.](#)
- d. [papilla.](#)
- e. [hilus.](#)

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11. The tip of each papilla, perforated by the openings of collecting ducts, is called the:

- a. [medullary ray.](#)
- b. [capsule.](#)
- c. [column of Bertini.](#)
- d. [area cribosa.](#)
- e. [hilus.](#)

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12. The space into which one medullary pyramid drains is called one:

- a. [column.](#)
- b. [papilla.](#)
- c. [calyx.](#)
- d. [pelvis.](#)
- e. [hilus.](#)

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13. The large space lined by transitional epithelium which funnels urine into the ureter is called the:

- a. [capsule.](#)
- b. [papillae.](#)
- c. [calyx.](#)
- d. [pelvis.](#)
- e. [hilus.](#)

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14. The capsule of the kidney consists of:

- a. [transitional epithelium.](#)
- b. [dense fibrous connective tissue.](#)
- c. [loose cellular connective tissue with many lymphocytes.](#)
- d. [adipose connective tissue.](#)
- e. [endothelium.](#)

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15. The region of the kidney where artery, vein and ureter are attached is called the:

- a. [cortex.](#)
- b. [medulla.](#)
- c. [calyx.](#)
- d. [pelvis.](#)
- e. [hilus.](#)

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16. The renal sinus, a space within the hilus that surrounds the renal pelvis, is filled with:
- extracellular fluid.
 - air.
 - loose adipose connective tissue.
 - urine.
 - dense fibrous connective tissue.
-
17. A principal function for the glomerulus is:
- filtration of blood from capillaries into urinary space.
 - active resorption of nutrients from ultrafiltrate back into the blood.
 - Volume reduction tubular fluid, with ions pumped from lumen into intersitium followed by diffusion of water.
 - regulated passive resorption of water through the medullary concentration gradient.
 - regulation of plasma volume and blood pressure through renin secretion.
-
18. A principle function for the proximal convoluted tubule is:
- filtration of blood from capillaries into urinary space.
 - active resorption of nutrients from ultrafiltrate back into the blood.
 - establishment of medullary concentration gradient.
 - regulated passive resorption of water through the medullary concentration gradient.
 - regulation of plasma volume and blood pressure through renin secretion.
-
19. A principal function for the juxtaglomerular apparatus is:
- filtration of blood from capillaries into the urinary space.
 - active resorption of nutrients from ultrafiltrate back into the blood.
 - volume reduction of tubular fluid, with ions pumped from lumen into intersitium followed by diffusion of water.
 - regulated passive resorption of water through the medullary concentration gradient.
 - regulation of plasma volume and blood pressure through renin secretion.
-
20. A principal function for the loop of Henle is:
- filtration of blood from capillaries into urinary space.
 - active resorption of nutrients from ultrafiltrate back into the blood.
 - establishment of the medullary concentration gradient.
 - regulated passive resorption of water through the medullary concentration gradient.
 - regulation of plasma volume and blood pressure through renin secretion.
-

21. A principal function for the distal convoluted tubule is:
- establishment of the medullary concentration gradient.
 - active resorption of nutrients from ultrafiltrate back into the blood.
 - volume reduction of tubular fluid, with ions pumped from lumen into interstitium followed by diffusion of water.
 - regulated passive resorption of water through the medullary concentration gradient.
 - regulation of plasma volume and blood pressure through renin secretion.
-
22. Fluid which is transported from the lumen of the proximal and distal convoluted tubules into the cortical interstitium is take up by the:
- collecting ducts.
 - pelvis.
 - podocytes.
 - peritubular capillaries.
 - loops of Henle.
-
23. A principle function for the collecting duct is:
- filtration of blood from capillaries into urinary space.
 - active resorption of nutrients from ultrafiltrate back into the blood.
 - volume reduction tubular fluid, with ions pumped from lumen into interstitium followed by diffusion of water.
 - regulated passive resorption of water through the medullary concentration gradient.
 - regulation of plasma volume and blood pressure through rennin secretion.
-
24. The region of glomerulus (renal corpuscle) where the capillary tuft is attached is called the:
- mesangium.
 - area cribosa.
 - vascular pole.
 - urinary (tubular) pole.
 - juxtaglomerular apparatus.
-
25. The region of glomerulus (renal corpuscle) where the urinary space drains into the proximal convoluted tubule is called the:
- mesangium.
 - area cribosa.
 - vascular pole.
 - urinary (tubular) pole.

- e. [juxtaglomerular apparatus.](#)

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26. The lumen of glomerular capillaries is lined by:

- a. [mesangial cells.](#)
- b. [podocytes foot processes.](#)
- c. [juxtaglomerular cells.](#)
- d. [continuous endothelium.](#)
- e. [fenestrated endothelium.](#)

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27. Interstitial cells located in matrix between glomerular capillaries are called:

- a. [pericytes.](#)
- b. [juxtaglomerular cells.](#)
- c. [podocytes.](#)
- d. [mesangial cells.](#)
- e. [fibroblasts.](#)

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28. Glomerular capillaries are supported on the outside (adjacent to the urinary space) by:

- a. [mesangial cells.](#)
- b. [fibroblasts.](#)
- c. [podocyte foot processes.](#)
- d. [pericytes.](#)
- e. [juxtaglomerular cells.](#)

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29. Smooth muscle cells of renal arterioles which are modified for endocrine secretion are called:

- a. [mesangial cells.](#)
- b. [fibroblasts.](#)
- c. [podocytes.](#)
- d. [pericytes.](#)
- e. [juxtaglomerular cells.](#)

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30. The outer lining of Bowman's capsule (the parietal epithelium) consists of a single layer of:

- a. [squamous cells.](#)
- b. [cuboidal cells.](#)
- c. [podocytes.](#)
- d. [pericytes.](#)
- e. [mesangial cells.](#)

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31. Filtration slits are located between adjacent:

- a. [layers of the glomerular basement membrane.](#)
- b. [capillary fenestrations.](#)
- c. [podocyte foot processes.](#)
- d. [mesangial cells.](#)
- e. [capillary endothelial cells.](#)

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32. Podocytes are:

- a. [simple squamous epithelial cells.](#)
- b. [simple cuboidal epithelial cells.](#)
- c. [simple columnar epithelial cells.](#)
- d. [unique epithelial cells with many interdigitating processes.](#)
- e. [not epithelial cells.](#)

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33. Mesangial cells are:

- a. [simple squamous epithelial cells.](#)
- b. [simple cuboidal epithelial cells.](#)
- c. [simple columnar epithelial cells.](#)
- d. [unique epithelial cells with many interdigitating processes.](#)
- e. [not epithelial cells.](#)

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34. The normal thickness of the glomerular basement membrane is between:

- a. [30 and 35 nanometers.](#)
- b. [0.3 and 0.35 micrometers.](#)
- c. [3 and 3.5 micrometers.](#)
- d. [30 and 35 micrometers.](#)
- e. [300 and 350 micrometers.](#)

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35. A basement membrane is also found beneath which of the following epithelia?

- a. [bowman's capsule.](#)
- b. [proximal convoluted tubule.](#)
- c. [distal convoluted tubule.](#)
- d. [macula densa.](#)
- e. [all of the above.](#)

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36. The component of the glomerular filtration barrier which is permeable to water and small molecules but not larger molecules like proteins is the:

- a. [mesangial cells.](#)
- b. [podocyte foot processes.](#)
- c. [fenestrated endothelium.](#)
- d. [glomerular basement membrane.](#)

- e. [bowman's capsule.](#)

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37. The component of the glomerular filtration barrier which holds back blood cells but allows fluid and large molecules to pass freely is the:

- a. [mesangial cells.](#)
- b. [podocyte foot processes.](#)
- c. [fenestrated endothelium.](#)
- d. [glomerular basement membrane.](#)
- e. [bowman's capsule.](#)

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38. Extraglomerular mesangial cells (laci cells) are located:

- a. [at the vascular pole of the glomerulus.](#)
- b. [at the urinary pole of the glomerulus.](#)
- c. [within the macula densa.](#)
- d. [encircling the proximal tubule.](#)
- e. [between adjacent podocyte foot processes.](#)

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39. Epithelial cells lining the proximal convoluted tubule:

- a. [have an apical brush border of microvilli for active absorption from the lumen.](#)
- b. [have basal infoldings and interdigitations with adjacent cells for increased basal membrane surface area.](#)
- c. [have many mitochondria to provide energy for active transport against a concentration gradient .](#)
- d. [are cuboidal.](#)
- e. [all of the above.](#)

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40. Epithelial cells lining the distal convoluted tubule:

- a. [have an apical brush border of microvilli for active absorption from the lumen.](#)
- b. [have basal infoldings and interdigitations with adjacent cells for increased basal membrane surface area.](#)
- c. [have many mitochondria to provide energy for active transport against a concentration gradient.](#)
- d. [are cuboidal.](#)
- e. [all of the above *except* the brush border.](#)

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41. The proximal convoluted tubule is:

- a. [several times longer than the distal convoluted tubule.](#)
- b. [about the same length as the distal convoluted tubule.](#)
- c. [less than half the length of the distal convoluted tubule.](#)
- d. [much narrower than the distal convoluted tubule.](#)

- e. [any of the above may be true, depending on location in cortex.](#)

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42. Epithelial cells lining the thin limb of the loop of Henle:

- a. [have an apical brush border of microvilli for active absorption from the lumen.](#)
- b. [contain numerous basally located secretory vesicles.](#)
- c. [contain any ribosomes and rough endoplasmic reticulum for extensive protein synthesis.](#)
- d. [are simple squamous.](#)
- e. [all of the above.](#)

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43. Epithelial cells lining collecting ducts:

- a. [stain less intensely with H & E than those lining proximal or distal tubules.](#)
- b. [often display distinct intercellular boundaries, unlike those lining proximal or distal tubules.](#)
- c. [regulate their permeability to water in response to antidiuretic hormone concentration.](#)
- d. [are simple cuboidal.](#)
- e. [all of the above.](#)

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44. The macula densa consists of:

- a. [extraglomerular mesangial cells.](#)
- b. [closely-packed nuclei in the epithelium of the distal convoluted tubule.](#)
- c. [modified smooth muscle cells.](#)
- d. [closely-packed nuclei in the epithelium of the proximal convoluted tubule.](#)
- e. [densely packed collagen fibers in the juxtaglomerular interstitium.](#)

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45. Juxtaglomerular cells secrete:

- a. [urea.](#)
- b. [antidiuretic hormone.](#)
- c. [renin.](#)
- d. [angiotensinogen.](#)
- e. [aldosterone.](#)

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46. Arteries which form arches along the border between the cortex and the medulla of the kidney are called:

- a. [interlobar arteries.](#)
- b. [interlobular arteries.](#)
- c. [arcuate arteries.](#)
- d. [afferent arteries.](#)

- e. [vasa recta.](#)

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47. Arteries which extend into the renal cortex, usually perpendicular to its surface, are called:

- a. [interlobar arteries.](#)
- b. [interlobular arteries.](#)
- c. [arcuate arteries.](#)
- d. [afferent arteries.](#)
- e. [vasa recta.](#)

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48. Bundles of parallel vessels which extend into the medulla are called:

- a. [interlobar arteries.](#)
- b. [interlobular arteries.](#)
- c. [arcuate arteries.](#)
- d. [afferent arteries.](#)
- e. [vasa recta.](#)

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49. Vessels which connect interlobular arteries with glomerular capillaries are called:

- a. [arcuate arteries.](#)
- b. [afferent arterioles.](#)
- c. [efferent arterioles.](#)
- d. [peritubular capillaries.](#)
- e. [vasa recta.](#)

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50. Vessels which connect glomerular capillaries with peritubular capillaries are called:

- a. [arcuate arteries.](#)
- b. [afferent arterioles.](#)
- c. [efferent arterioles.](#)
- d. [peritubular capillaries.](#)
- e. [vasa recta.](#)

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51. Vessels which surround proximal and distal convoluted tubules, receiving reabsorbed water and nutrients, are called:

- a. [arcuate arteries.](#)
- b. [afferent arterioles.](#)
- c. [efferent arterioles.](#)
- d. [peritubular capillaries.](#)
- e. [vasa recta.](#)

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52. The volume of blood decreases while passing through the:

- a. [interlobular arteries.](#)
 - b. [afferent arterioles.](#)
 - c. [glomerular capillaries.](#)
 - d. [efferent arterioles.](#)
 - e. [peritubular capillaries.](#)
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53. The volume of blood increases while passing through the:
- a. [interlobular arteries.](#)
 - b. [afferent arterioles.](#)
 - c. [glomerular capillaries.](#)
 - d. [efferent arterioles.](#)
 - e. [peritubular capillaries.](#)
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54. Vessels which form a counter-current exchange, to maintain an osmotic gradient established by long loops of Henle, are called:
- a. [arcuate arteries.](#)
 - b. [afferent arterioles.](#)
 - c. [efferent arterioles.](#)
 - d. [peritubular capillaries.](#)
 - e. [vasa recta.](#)
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55. Interstitial fluid in the kidney remains essentially iso-osmotic with blood in each of the following regions EXCEPT:
- a. [cortex, surrounding proximal convoluted tubules.](#)
 - b. [cortex, surrounding distal convoluted tubules.](#)
 - c. [cortex, within the glomerular mesangium.](#)
 - d. [cortex, surrounding Bowman's capsule.](#)
 - e. [medulla, surrounding loops of Henle.](#)
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56. Interstitial oxygen concentration is lowest in which of the following regions?
- a. [outer cortex.](#)
 - b. [juxtamedullary cortex.](#)
 - c. [outer medulla.](#)
 - d. [inner medulla, in the papillae.](#)
 - e. [interstitial oxygen concentration is similar in all the above areas.](#)
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57. The essential function for the medullary concentration gradient is to permit the:
- a. [passive resorption of nutrients from tubular fluid.](#)
 - b. [active resorption of ions from tubular fluid.](#)
 - c. [passive resorption of ions from tubular fluid.](#)
 - d. [active resorption of water from tubular fluid.](#)

- e. [passive resorption of water from tubular fluid.](#)

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58. The volume of tubular fluid decreases as it passes through which of the following?

- a. [proximal convoluted tubule.](#)
- b. [loop of Henle.](#)
- c. [distal convoluted tubule.](#)
- d. [collecting duct.](#)
- e. [all of the above.](#)

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59. Which segment of the nephron functions to concentrate tubular fluid by passive resorption of water as it as it passes through an osmotic concentration gradient?

- a. [glomerulus.](#)
- b. [proximal convoluted tubule.](#)
- c. [loop of Henle.](#)
- d. [distal convoluted tubule.](#)
- e. [collecting duct.](#)

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60. Which segment of the nephron functions to filter fluid from blood into urinary space?

- a. [glomerulus.](#)
- b. [proximal convoluted tubule.](#)
- c. [loop of Henle.](#)
- d. [distal convoluted tubule.](#)
- e. [collecting duct.](#)

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61. Which segment of the nephron functions to reduce tubular fluid volume by pumping ions (which are followed by water) into the interstitium?

- a. [glomerular capillaries.](#)
- b. [glomerular mesangium.](#)
- c. [juxtaglomerular apparatus.](#)
- d. [distal convoluted tubule.](#)
- e. [collecting duct.](#)

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62. Which segment of the nephron functions to establish a medullary concentration gradient?

- a. [glomerulus.](#)
- b. [proximal convoluted tubule.](#)
- c. [loop of Henle.](#)
- d. [distal convoluted tubule.](#)
- e. [collecting duct.](#)

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63. Which segment of the nephron functions by active resorption of nutrients from ultrafiltrate back into blood?
- a. [glomerulus.](#)
 - b. [proximal convoluted tubule.](#)
 - c. [loop of Henle.](#)
 - d. [juxtaglomerular apparatus.](#)
 - e. [collecting duct.](#)
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64. Which segment of the nephron functions to regulate blood pressure through renin secretion?
- a. [glomerulus.](#)
 - b. [proximal convoluted tubule.](#)
 - c. [juxtaglomerular apparatus.](#)
 - d. [loop of Henle.](#)
 - e. [collecting duct.](#)
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65. Which cells respond to aldosterone?
- a. [podocytes.](#)
 - b. [epithelial cells of distal tubules.](#)
 - c. [juxtaglomerular cells.](#)
 - d. [epithelial cells of Bowman's capsule.](#)
 - e. [mesangial cells.](#)
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66. Which cells respond to antidiuretic hormone (ADH)?
- a. [podocytes.](#)
 - b. [mesangial cells.](#)
 - c. [juxtaglomerular cells.](#)
 - d. [epithelial cells of Bowman's capsule.](#)
 - e. [epithelial cells of collecting duct.](#)
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67. Which cells secrete a product which converts angiotensinogen to angiotensin?
- a. [podocytes.](#)
 - b. [epithelial cells of proximal and distal convoluted tubules.](#)
 - c. [juxtaglomerular cells.](#)
 - d. [epithelial cells of Bowman's capsule.](#)
 - e. [epithelial cells of collecting duct.](#)
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68. Which cells secrete angiotensinogen?
- a. [podocytes](#)
 - b. [epithelial cells of proximal and distal convoluted tubules.](#)
 - c. [juxtaglomerular cells.](#)

- d. [hepatocytes.](#)
- e. [epithelial cells of collecting duct.](#)

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69. The volume of tubular fluid is greatest as it leaves which of the following?

- a. [Bowman's space.](#)
- b. [proximal convoluted tubule.](#)
- c. [loop of Henle.](#)
- d. [distal convoluted tubule.](#)
- e. [collecting duct.](#)

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70. Which epithelial cells of the urinary tract are specialized with an apical brush border of microvilli?

- a. [Bowman's capsule epithelium.](#)
- b. [proximal convoluted tubule epithelium.](#)
- c. [epithelium lining the thin segment of the loop of Henle.](#)
- d. [distal convoluted tubule epithelium.](#)
- e. [collecting duct epithelium.](#)

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71. Which epithelial cells of the urinary tract are specialized with basal membrane folds closely associated with many mitochondria?

- a. [Bowman's capsule epithelium.](#)
- b. [proximal and distal convoluted tubule epithelium.](#)
- c. [epithelium lining the thin segment of the loop of Henle.](#)
- d. [collecting duct epithelium.](#)
- e. [transitional epithelium.](#)

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72. Which of the following epithelial cells of the urinary tract are simple squamous, without apparent structural specialization?

- a. [Bowman's capsule epithelium.](#)
- b. [glomerular epithelium \(podocytes\).](#)
- c. [transitional epithelium.](#)
- d. [distal convoluted tubule epithelium.](#)
- e. [collecting duct epithelium.](#)

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73. Which epithelial cells of the urinary tract are specialized with folds of thickened plasma membrane, and including many apical vesicles lined by similar membrane, which may permit rapid expansion of the cells' surface when stretched?

- a. [bowman's capsule epithelium.](#)
- b. [glomerular epithelium \(podocytes\).](#)
- c. [epithelium lining the thin segment of the loop of Henle.](#)
- d. [distal convoluted tubule epithelium.](#)

- e. [transitional epithelium.](#)
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74. Interstitial cells within the renal glomerulus, whose functions are uncertain but may include phagocytosis and maintenance of the glomerular basement membrane, are the:
- a. [podocytes.](#)
 - b. [macula densa.](#)
 - c. [juxtaglomerular cells.](#)
 - d. [mesangial cells.](#)
 - e. [transitional epithelium.](#)
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75. Transitional epithelium lines all of the following EXCEPT the:
- a. [renal pelvis.](#)
 - b. [ureter.](#)
 - c. [bladder.](#)
 - d. [proximal urethra.](#)
 - e. [Bowman's capsule.](#)
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